Chapter 25 Atoms

'If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words?'

Feynman’s question

Feynman’s answer: All things are made of atoms.

- Democritus (460BC-370BC) atomos ----uncuttable

(Aristotle Air Fire Earth Water

- Dalton's Atomic Theory

1. All matter is made up of indivisible and indestructible particles called atoms.
2. All atoms of a given element are identical, both in mass and in properties.
3. Compounds are formed when atoms of different elements combine in the ratio of small whole numbers.
4. Elements and compounds are composed of definite arrangements of atoms, and chemical change occurs when the atomic arrays are rearranged.

•J. J. Thomson	plum pudding model
•Earnest Rutherford	nuclear model
•Niels Bohr	Bohr's model
•Erwin Schrödinger	Schrödinger's model ('electron clouds')

25.1 Hydrogen atom

The Hamiltonian for hydrogen-like atom is

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze_s^2}{r}$$

reduced mass

$$e_s^2 \equiv \frac{1}{4\pi\varepsilon_0} e^2$$

$Z = 1$, the equation is for hydrogen.

$Z \neq 1$, the equation applies to ions (He^+, Li^{++})

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

In spherical coordinate system

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

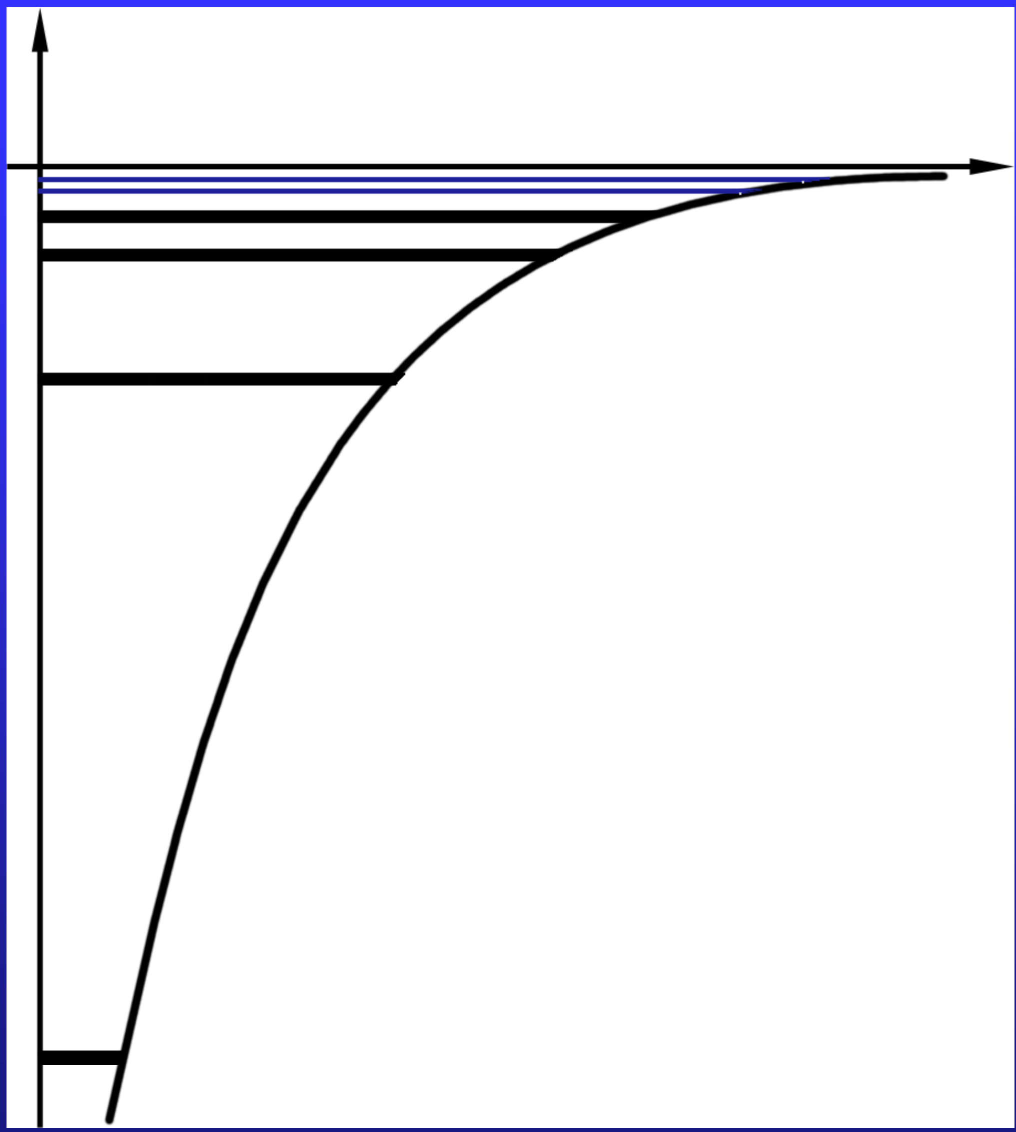
The angular part of the operator

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \equiv -\frac{\hat{L}^2}{\hbar^2}$$

The Schrödinger equation is

$$-\frac{\hbar^2}{2\mu}\underline{\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{1}{r^2}\frac{\hat{L}^2}{\hbar^2}\right]}\psi - \frac{Ze_s^2}{r}\psi = E\psi$$

$E < 0$ for bound states



$$\psi(r,\theta,\varphi) = R(r)Y(\theta,\varphi)$$

Radial part

angular part

$$\hat{L}^2\, Y_{\ell m}(\theta,\varphi) = \ell(\ell+1)\hbar^2\, Y_{\ell m}(\theta,\varphi)$$

integer
ℓ=1,2,3,…

the square of the angular momentum operator(section 25.4)

spherical harmonic function
$m = -\ell, -\ell+1, \ldots, \ell$
(Appendix K)

The radial part of equation is

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R}{\partial r}\right)+\left[\frac{2\mu}{\hbar^2}\left(E+\frac{Ze_s^2}{r}\right)-\frac{\ell(\ell+1)}{r^2}\right]R=0$$

The eigenfunctions and eigenvalues of above equation are

$$R_{n\ell}(r) \quad \text{and} \quad E_n=-\frac{\mu Z^2 e_s^4}{2\hbar^2}\cdot\frac{1}{n^2}\ , \quad n=1,2,\cdots$$

Therefore, the total wave functions are

$$\begin{cases}\psi_{n\ell m}=R_{n\ell}(r)Y_{\ell m}(\theta,\varphi) \\ E_n=-\dfrac{\mu Z^2 e_s^4}{2\hbar^2}\cdot\dfrac{1}{n^2}\ , \quad n=1,2,\cdots \\ \ell=0,\ 1,\ 2,\ \cdots,\ n-1 \\ m=-\ell,\ -(\ell-1),\cdots,(\ell-1),\ell\end{cases}$$

n principal quantum number
ℓ angular (azimuthal) quantum number

m magnetic quantum number

(n, ℓ, m) – quantum state, energy ~ (n)
same n (energy), different ℓ, m – degenerate

简并

For hydrogen, degeneracy is

$$\sum_{\ell=0}^{n-1}\sum_{m=-\ell}^{\ell} 1 = \sum_{\ell=0}^{n-1}(2\ell+1) = n^2$$

$$L^2 = \ell(\ell+1)\hbar^2 \le (n-1)n\hbar^2 < (n\hbar)^2$$

25.2 Energy level and transition

$$h\nu = E_a - E_b$$

$\ell = 0$, the line is called sharp series, 锐

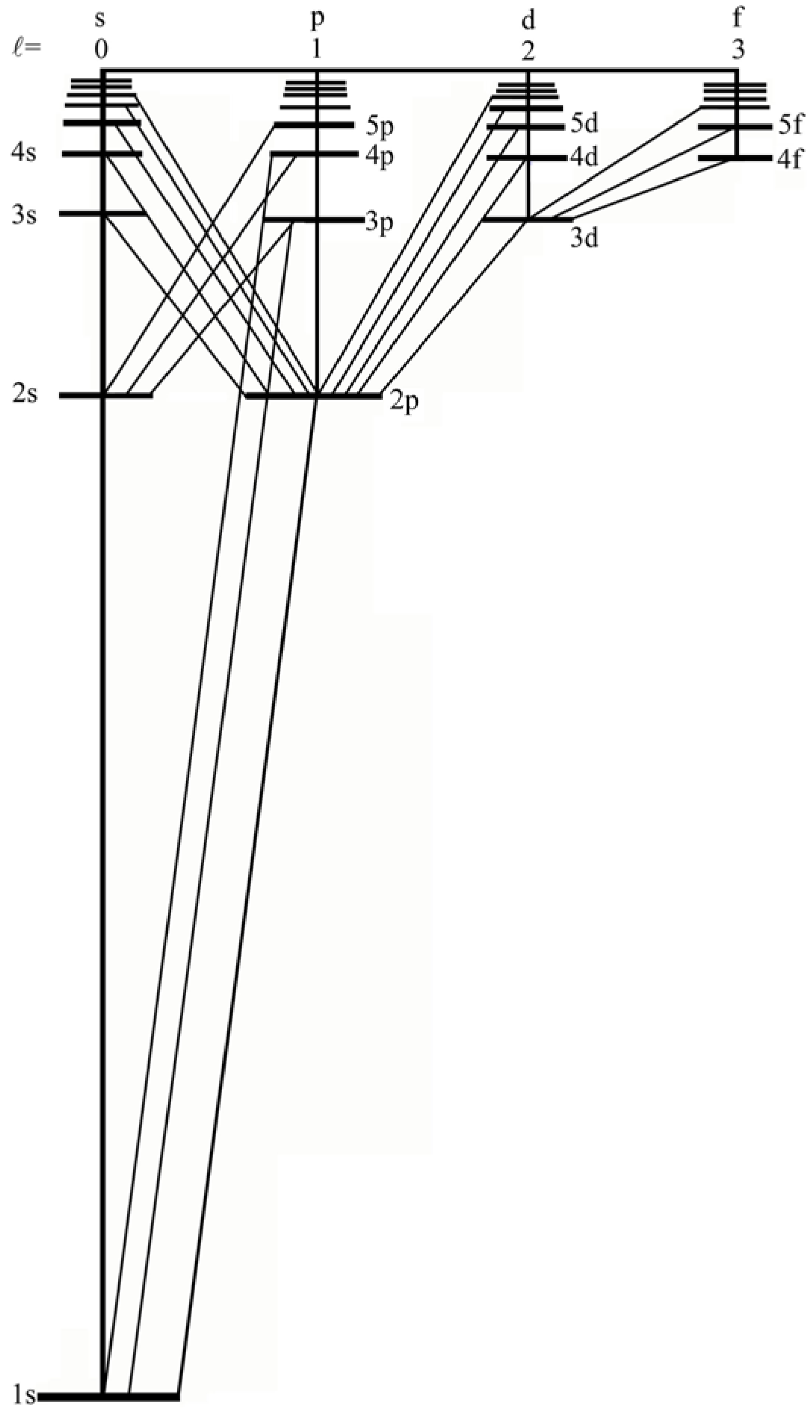
$\ell = 1$ principal series, 主

$\ell = 2$ diffuse series, and 漫

$\ell = 3$ fundamental series etc. 基

selection rules and most important one

$$\Delta\ell = \pm 1$$



s
p
d
f
ℓ=
0
1
2
3
5p
4p
3p
2p
5d
4d
3d
5f
4f
4s
3s
2s
1s

$$E_n = -\frac{m_e Z^2 e_s^4}{\left(1+\frac{m_e}{m_p}\right)\cdot 2\hbar^2}\cdot\frac{1}{n^2}$$

$$= -\frac{Z^2 e_s^2}{2\eta\, a_0}\frac{1}{n^2}$$

For hydrogen, $\eta = 1.000\ 54$.
Singly ionized helium He^{+}, $Z = 2$, $\eta = 1.000\ 14$.
Deuterium, $\eta = 1.000\ 27$

Example 25.1 muonic hydrogen atom

$$m_\mu = 207 m_e$$

$$\mu_e = \frac{m_e m_p}{m_e + m_p}; \mu_\mu = \frac{207 m_e m_p}{207 m_e + m_p} = 186 m_e$$

$$\eta = \frac{1}{186}; \quad \langle r \rangle_1 = \eta\, a_0 = \frac{a_0}{186}; \quad E_1 = 186 E_{1\mathrm{H}}$$

to gain the information about nuclei,
to form muonic molecules to cause cold fusion,
to produce X rays of special frequency

25.3 Probability density

$$\psi_{n\ell m} = R_{n\ell}(r) Y_{\ell m}(\theta, \varphi)$$

The probability in the volume element $r^2 \sin\theta d\theta d\varphi dr$

$$|\psi_{n\ell m}|^2 r^2 \underline{\sin\theta d\theta d\varphi} dr = |\psi_{n\ell m}|^2 r^2 dr d\underline{\Omega}$$

Radial probability density

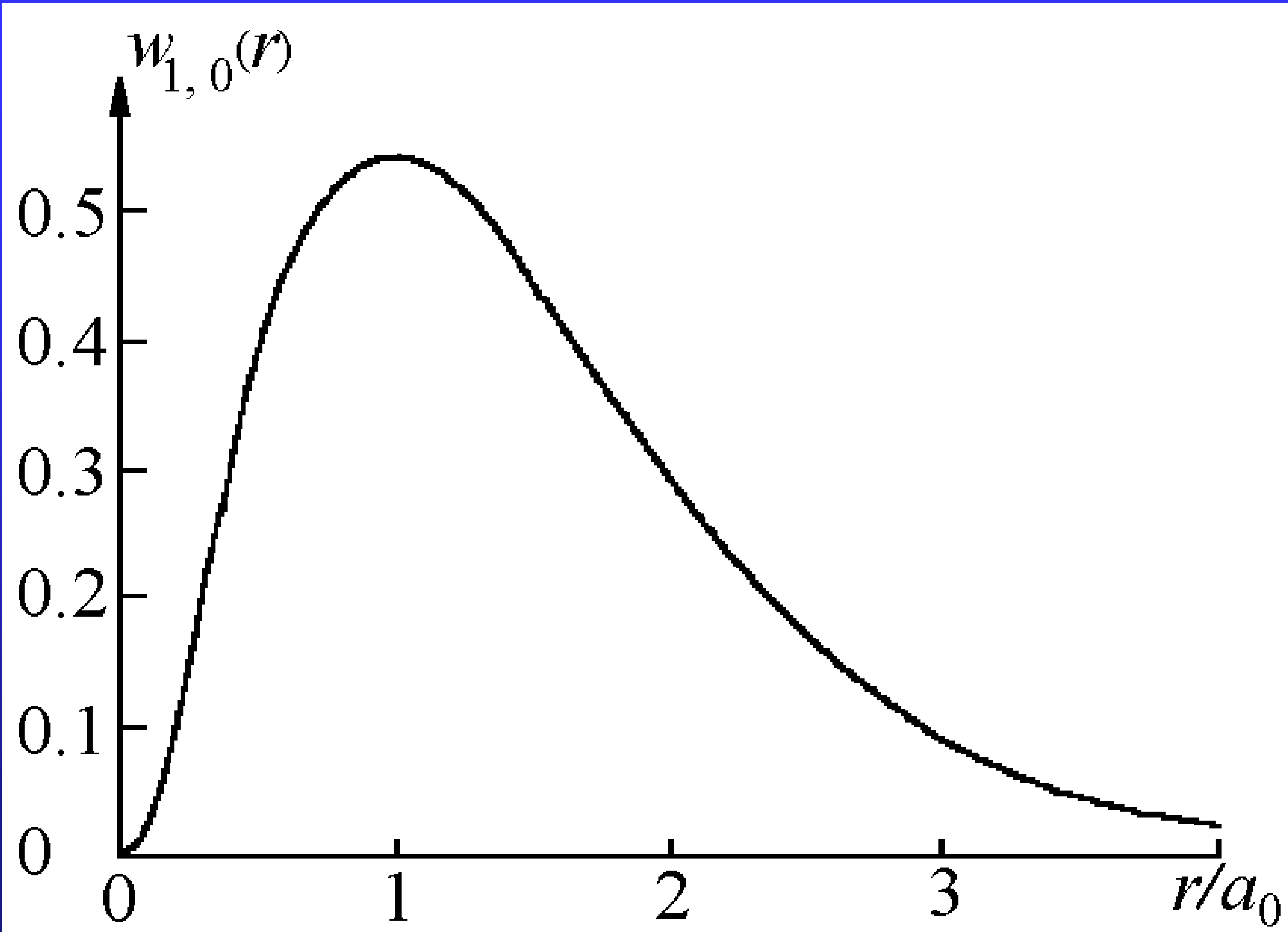
$$\int |Y_{\ell m}(\theta, \varphi)|^2 d\Omega = 1$$

$$w_{n\ell}(r) dr = r^2 dr \int d\Omega |\psi_{n\ell m}|^2$$

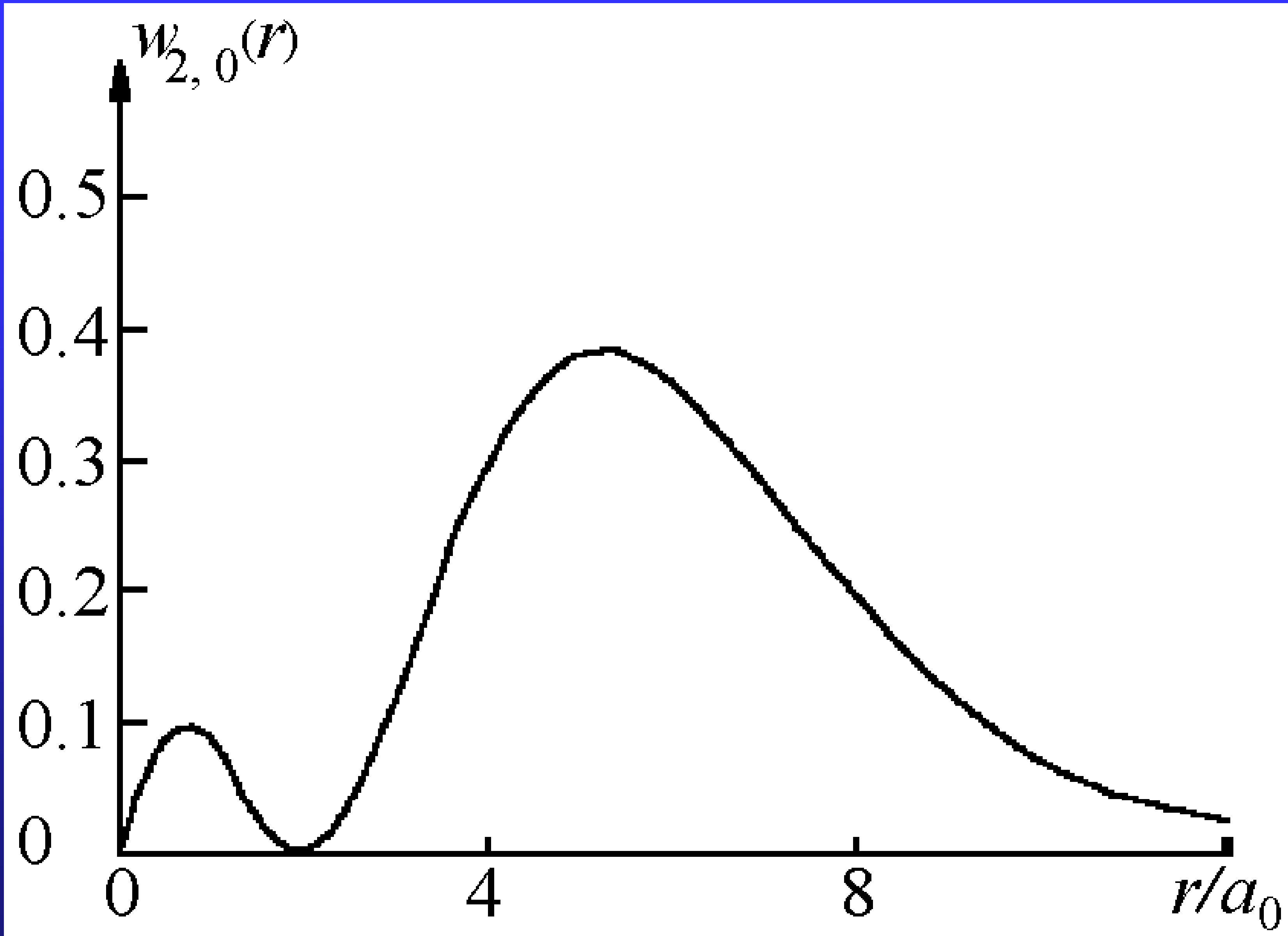
$$= R_{n\ell}^2(r)\, r^2 dr$$

$$R_{n\ell} \sim e^{-\frac{1}{2}\rho} \rho^{\ell} \mathrm{L}_{n-\ell-1}^{2\ell+1}(\rho); \quad \mathrm{L}_0^{2n-1} = 1$$

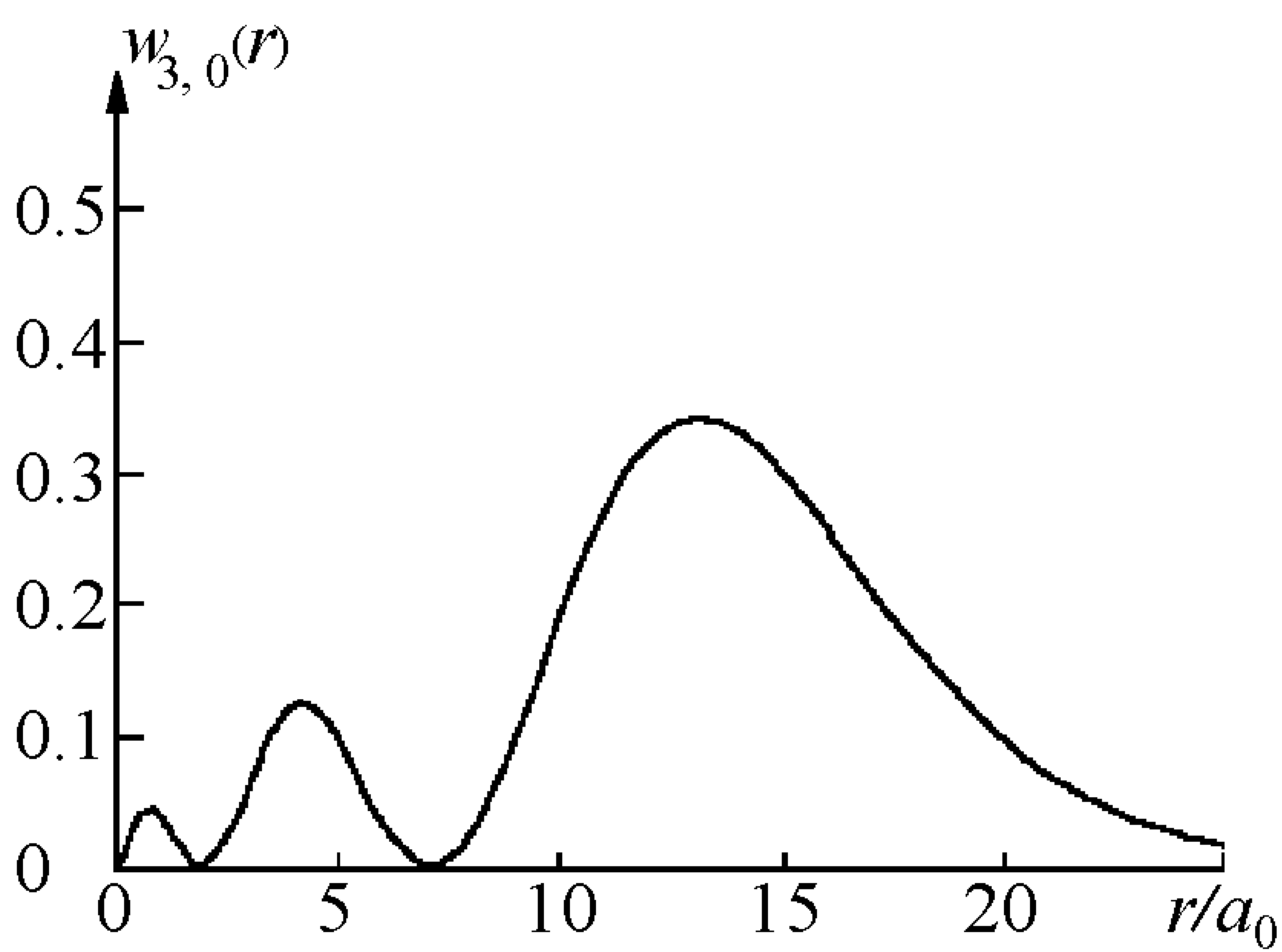
associated Laguerre polynomial

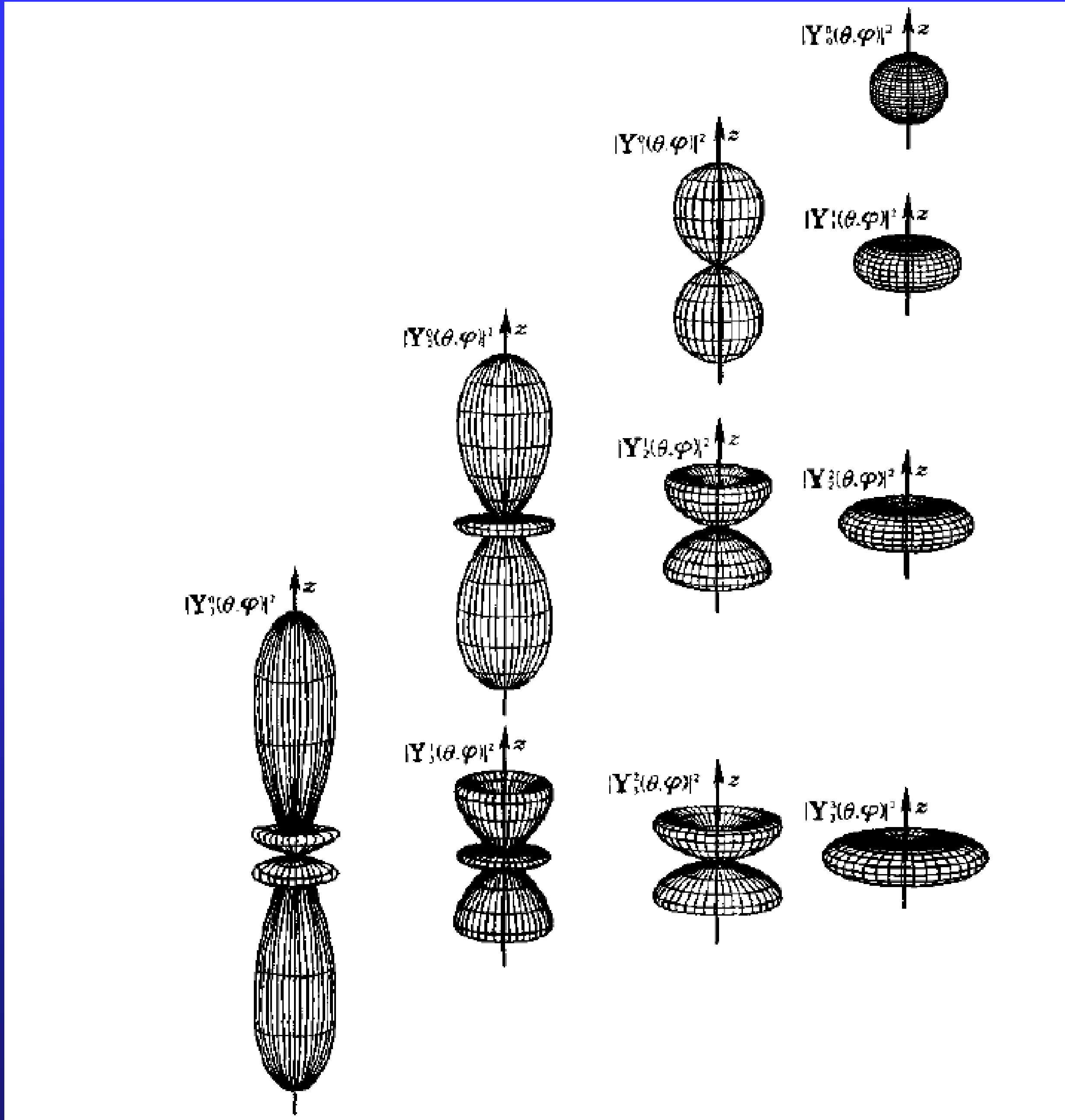


$w_{1,0}(r)$
0.5
0.4
0.3
0.2
0.1
0
0
1
2
3
r/a_0

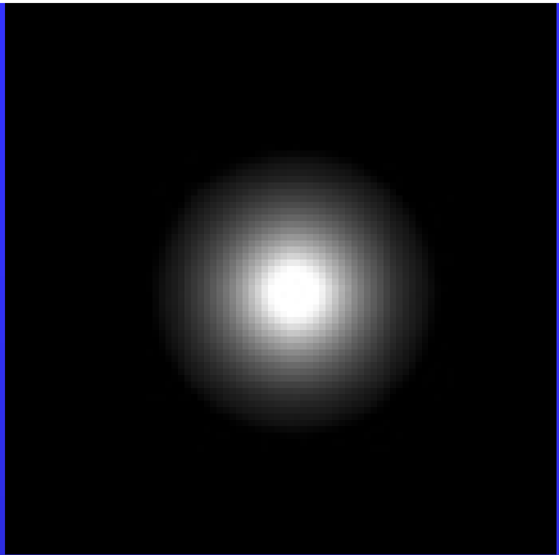


$w_{2,0}(r)$
0.5
0.4
0.3
0.2
0.1
0
0
4
8
r/a_0

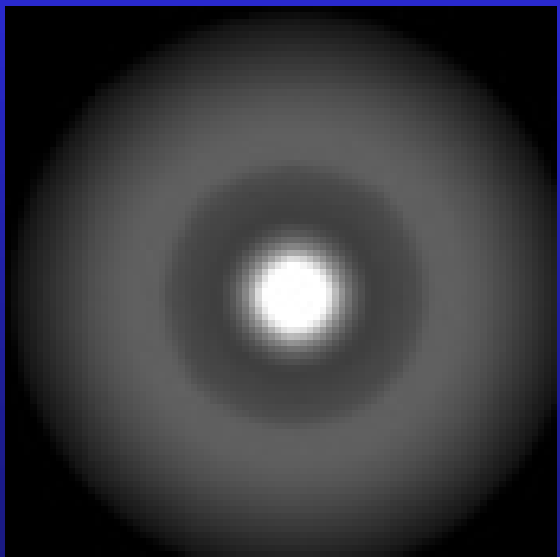

$w_{3,0}(r)$
0.5
0.4
0.3
0.2
0.1
0
0
5
10
15
20
r/a_0



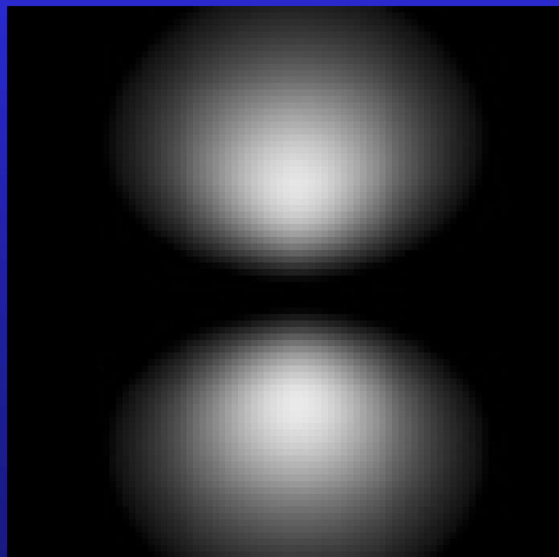
$|Y_0^0(\theta,\varphi)|^2$
z
$|Y_1^0(\theta,\varphi)|^2$
z
$|Y_1^1(\theta,\varphi)|^2$
z
$|Y_2^0(\theta,\varphi)|^2$
z
$|Y_2^1(\theta,\varphi)|^2$
z
$|Y_2^2(\theta,\varphi)|^2$
z
$|Y_3^0(\theta,\varphi)|^2$
z
$|Y_3^1(\theta,\varphi)|^2$
z
$|Y_3^2(\theta,\varphi)|^2$
z
$|Y_3^3(\theta,\varphi)|^2$
z



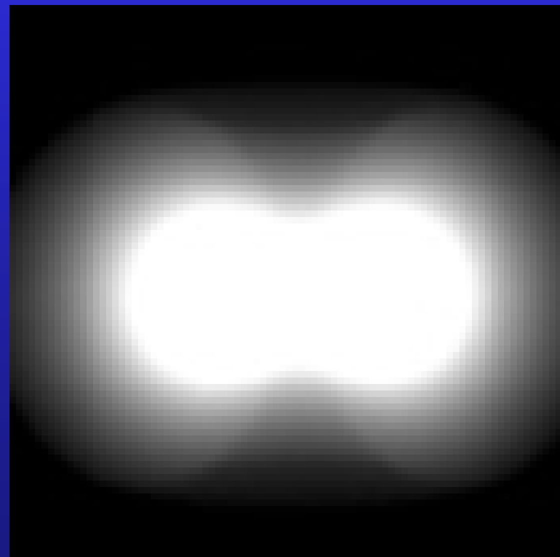
ψ_{100}



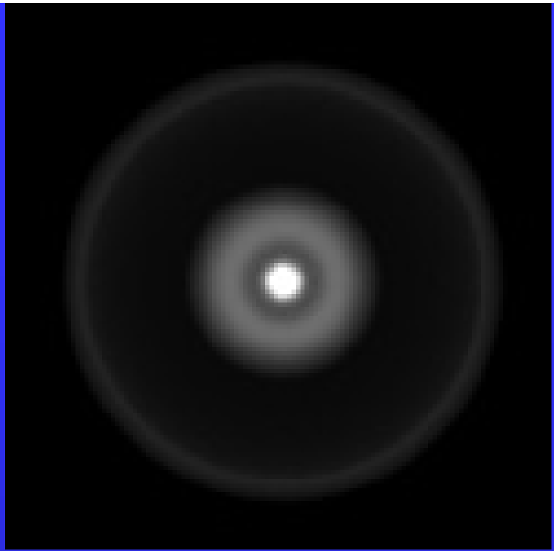
ψ_{200}



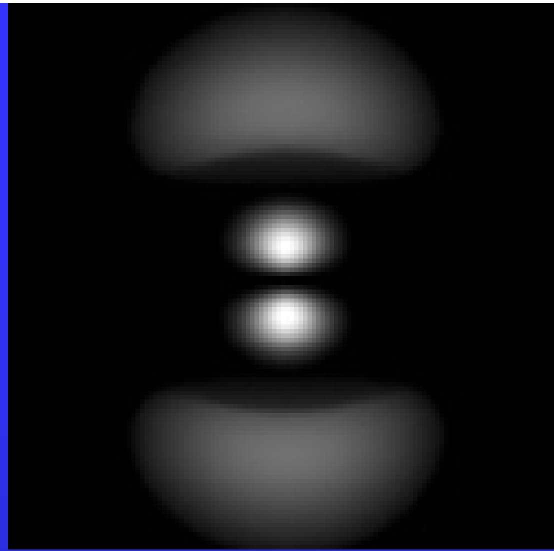
ψ_{210}



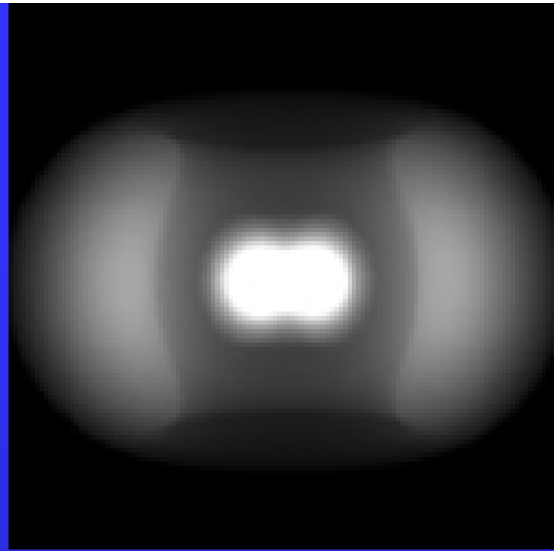
ψ_{211}



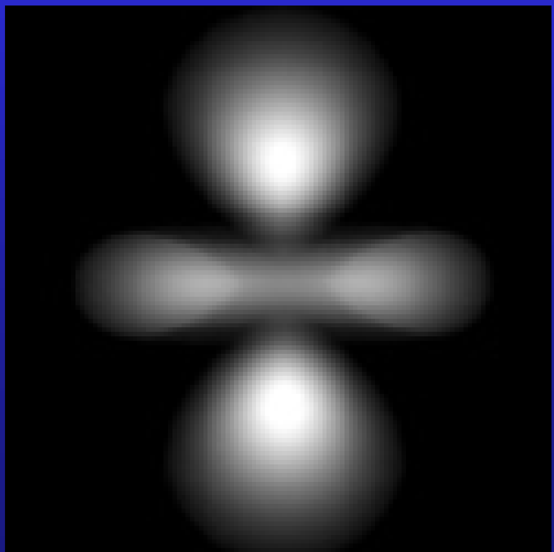
ψ_{300}



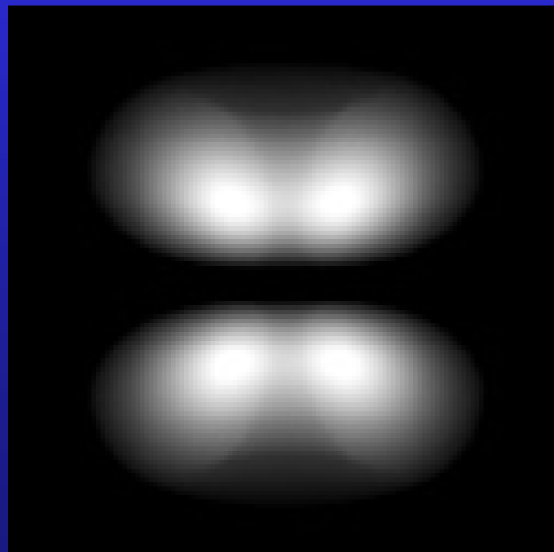
ψ_{310}



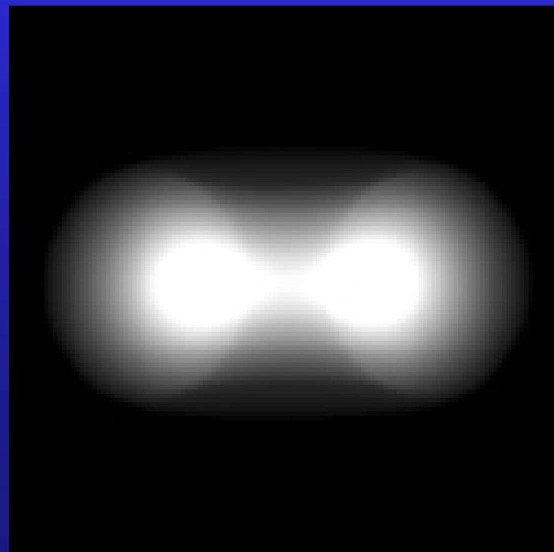
ψ_{311}



ψ_{320}



ψ_{321}



ψ_{322}

$$\langle r \rangle = \int_0^\infty r\, R_{n\ell}^2\, r^2 dr$$

$$= \frac{1}{2}\left[3\, n^2 - \ell(\ell+1)\right] a_0 \neq n^2 a_0$$

$$\frac{\partial}{\partial \rho}\left[\rho^2 R_{n,n-1}^2\right] = 0$$

$$n - 1 = \max(\ell)$$

$$r_{mp} = \frac{1}{Z} n^2 a_0$$

the probability current density

$$\boldsymbol{j} = \frac{\hbar}{2\mathrm{i}\mu}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right]$$

We can see that

$$\psi_{n\ell m}(r,\theta,\varphi) = \underline{\tilde{\psi}_{n\ell m}(r,\theta)}e^{im\varphi}$$

real

Using

$$\nabla = \boldsymbol{e}_x\frac{\partial}{\partial x} + \boldsymbol{e}_y\frac{\partial}{\partial y} + \boldsymbol{e}_z\frac{\partial}{\partial z}$$

$$= \boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \boldsymbol{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi},$$

we have

$$j_r = \frac{\hbar}{2i\mu}\left[\tilde{\psi}_{n\ell m}(r,\theta)e^{-im\varphi}\frac{\partial}{\partial r}(\tilde{\psi}_{n\ell m}(r,\theta)e^{im\varphi})\right.$$

$$\left. -\tilde{\psi}_{n\ell m}(r,\theta)e^{im\varphi}\frac{\partial}{\partial r}(\tilde{\psi}_{n\ell m}(r,\theta)e^{-im\varphi})\right] = 0,$$

$$j_\theta = \frac{\hbar}{2i\mu}\left[\tilde{\psi}_{n\ell m}(r,\theta)e^{-im\varphi}\frac{1}{r}\frac{\partial}{\partial \theta}(\tilde{\psi}_{n\ell m}(r,\theta)e^{im\varphi})\right.$$

$$\left. -\tilde{\psi}_{n\ell m}(r,\theta)e^{im\varphi}\frac{1}{r}\frac{\partial}{\partial \theta}(\tilde{\psi}_{n\ell m}(r,\theta)e^{-im\varphi})\right] = 0,$$

$$j_\varphi = \frac{m\hbar}{\mu}\frac{1}{r\sin\theta}\tilde{\psi}^2(r,\theta) \neq 0.$$

So electron cloud is in motion.

Assignment: 25.3, 25.5

25.4 Angular momentum and spin

$$\hat{L} = \boldsymbol{r} \times \hat{\boldsymbol{p}} = \begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ x & y & z \\ \hat{p}_x & \hat{p}_y & \hat{p}_z \end{vmatrix}$$

$$= \boldsymbol{e}_x \left(y\hat{p}_z - z\hat{p}_y \right) + \boldsymbol{e}_y \left(z\hat{p}_x - x\hat{p}_z \right) + \boldsymbol{e}_z \left(x\hat{p}_y - y\hat{p}_x \right)$$

$$= e_x \frac{\hbar}{i} \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) + e_y \frac{\hbar}{i} \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) + e_z \frac{\hbar}{i} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

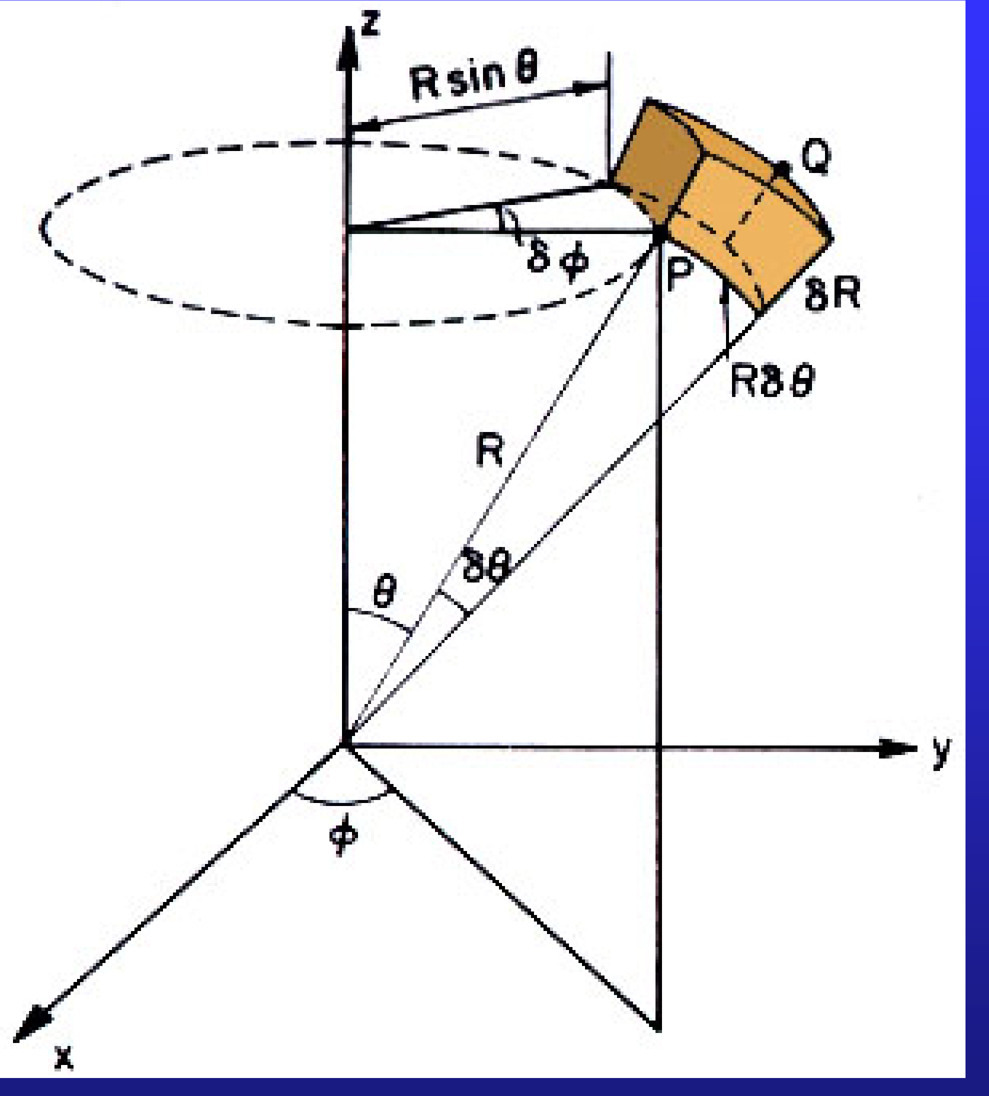
$$\hat{L}_x = \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \right)$$

$$\hat{L}_y = \frac{\hbar}{i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} \right)$$

$$\hat{L}_z = \frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right)$$

cyclic permutation

Spherical coordinates



The volume element, area element and the solid angle element are

$$d\tau = rd\theta r\sin\theta d\varphi dr = r^2 \sin\theta drd\theta d\varphi$$

$$dS = rd\theta r\sin\theta d\varphi = r^2 \sin\theta d\theta d\varphi$$

$$d\Omega = \frac{dS}{r^2} = \sin\theta d\theta d\varphi$$

Transformation between spherical coordinates and Cartesian coordinates

$$x = r\sin\theta\cos\varphi \qquad r^2 = x^2 + y^2 + z^2$$

$$y = r\sin\theta\sin\varphi \qquad \cos\theta = \frac{z}{r}, \quad \tan\varphi = \frac{y}{x}$$

$$z = r\cos\theta$$

Transformation of the derivative operators

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x}\frac{\partial}{\partial \theta} + \frac{\partial \varphi}{\partial x}\frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial y} = \frac{\partial r}{\partial y}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y}\frac{\partial}{\partial \theta} + \frac{\partial \varphi}{\partial y}\frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial z} = \frac{\partial r}{\partial z}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial z}\frac{\partial}{\partial \theta} + \frac{\partial \varphi}{\partial z}\frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial r}=\frac{\partial x}{\partial r}\frac{\partial}{\partial x}+\frac{\partial y}{\partial r}\frac{\partial}{\partial y}+\frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial \theta}=\frac{\partial x}{\partial \theta}\frac{\partial}{\partial x}+\frac{\partial y}{\partial \theta}\frac{\partial}{\partial y}+\frac{\partial z}{\partial \theta}\frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial \varphi}=\frac{\partial x}{\partial \varphi}\frac{\partial}{\partial x}+\frac{\partial y}{\partial \varphi}\frac{\partial}{\partial y}+\frac{\partial z}{\partial \varphi}\frac{\partial}{\partial z}$$

$$\frac{\partial x}{\partial \varphi}=-r\sin\theta\sin\varphi=-y,$$

$$\frac{\partial y}{\partial \varphi}=x,\frac{\partial z}{\partial \varphi}=0.$$

$$\frac{\partial}{\partial \varphi}=-y\frac{\partial}{\partial x}+x\frac{\partial}{\partial y}$$

$$\boxed{\hat{L}_z=\frac{\hbar}{i}\left(x\frac{\partial}{\partial y}-y\frac{\partial}{\partial x}\right)=\frac{\hbar}{i}\frac{\partial}{\partial \varphi}}$$

The square of angular momentum operator

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$= -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right]$$

The eigenvalue equation for the square of angular momentum operator is

$$\hat{L}^2 Y(\theta,\varphi) = L^2 Y(\theta,\varphi).$$

Solving the differential equation, we have

$$L^2 = \ell(\ell+1)\hbar^2, \quad \ell = 0,1,2,3\cdots$$

$$Y_{\ell m}(\theta,\varphi) = \mathrm{P}_l^m(\cos\theta)e^{im\varphi}, \; m = -\ell, -\ell+1, \cdots, \ell-1, \ell.$$

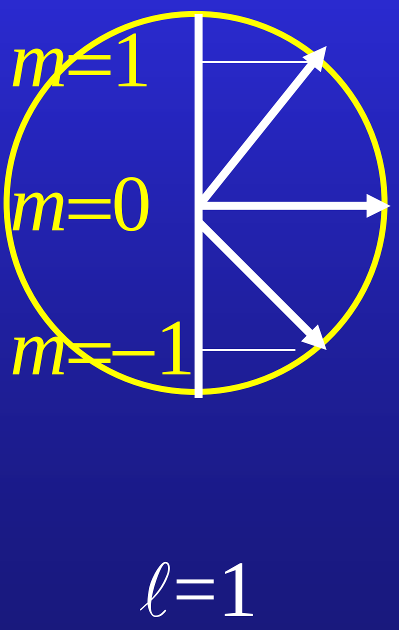
associated Legendre function

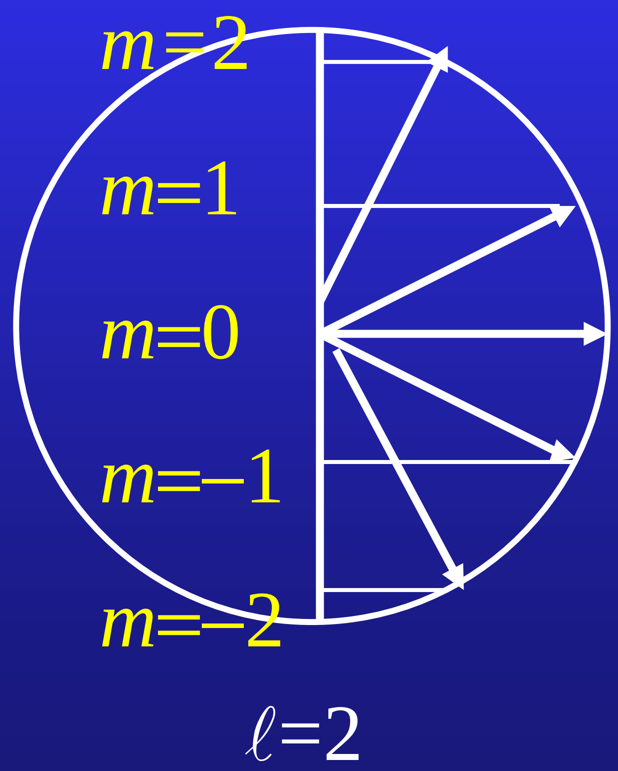
$$Y_{\ell m}(\theta,\varphi) \sim e^{im\varphi}$$

$$\hat{L}_z Y_{\ell m}(\theta,\varphi) = m\hbar Y_{\ell m}(\theta,\varphi)$$

$$\begin{cases} \hat{L}^2 Y_{\ell m}(\theta,\varphi) = \ell(\ell+1)\hbar^2 Y_{\ell m}(\theta,\varphi), \\ \hat{L}_z Y_{\ell m}(\theta,\varphi) = m\hbar Y_{\ell m}(\theta,\varphi). \end{cases}$$

$Y_{\ell m}(\theta,\varphi)$ are the common eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$.





Example 25.2 A rotator of the moment of inertia I is rotating around an axis. Find the energy levels and the eigen wavefunction of the system.

Solution: The Schrödinger equation is

$$\frac{L_z^2}{2I}$$

$$\boxed{-\frac{\hbar^2}{2I}\frac{\partial^2}{\partial\varphi^2}}\Phi(\varphi) = E\Phi(\varphi).$$

or

$$\frac{d^2}{d\varphi^2}\Phi(\varphi) + k^2\Phi(\varphi) = 0, \qquad k^2 = \frac{2IE}{\hbar^2}.$$

The general solution is

$$\Phi(\varphi) = Ce^{ik\varphi}.$$

Using the condition

$$\Phi(\varphi + 2\pi) = \Phi(\varphi)$$

or

$$Ce^{ik(\varphi+2\pi)} = Ce^{ik\varphi}, \qquad e^{i2\pi k} = 1$$

we have $k=m$, $m=0, \pm1, \pm2, \ldots$

$$\begin{cases} \Phi_m(\varphi) = \dfrac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}m\varphi} \\ E_m = \dfrac{m^2\hbar^2}{2I} \end{cases}$$

Electron spin

Lines in principal and sharp series are found all doublet,
in diffuse and fundamental series triplet.

well-known D line of sodium: D1 = 5896Å
D2 = 5890Å

fine structure

$2p \longrightarrow 1s$

In Section 19.1 $\mu_{\mathrm{L}} = I\pi r^2(-\boldsymbol{k})$

$$= \frac{ev}{2\pi r}\pi r^2(-\boldsymbol{k})$$

classical

$L = mrv$

$$= \frac{e}{2m}\cdot n\hbar(-\boldsymbol{k})$$

Bohr

$$\gamma = -\frac{e}{2m}$$

quantum

$$L_z = m\hbar$$

the gyromagnetic ratio

In a non-uniform magnetic field

$$\boldsymbol{F} = \mu \cdot \nabla \boldsymbol{B}$$

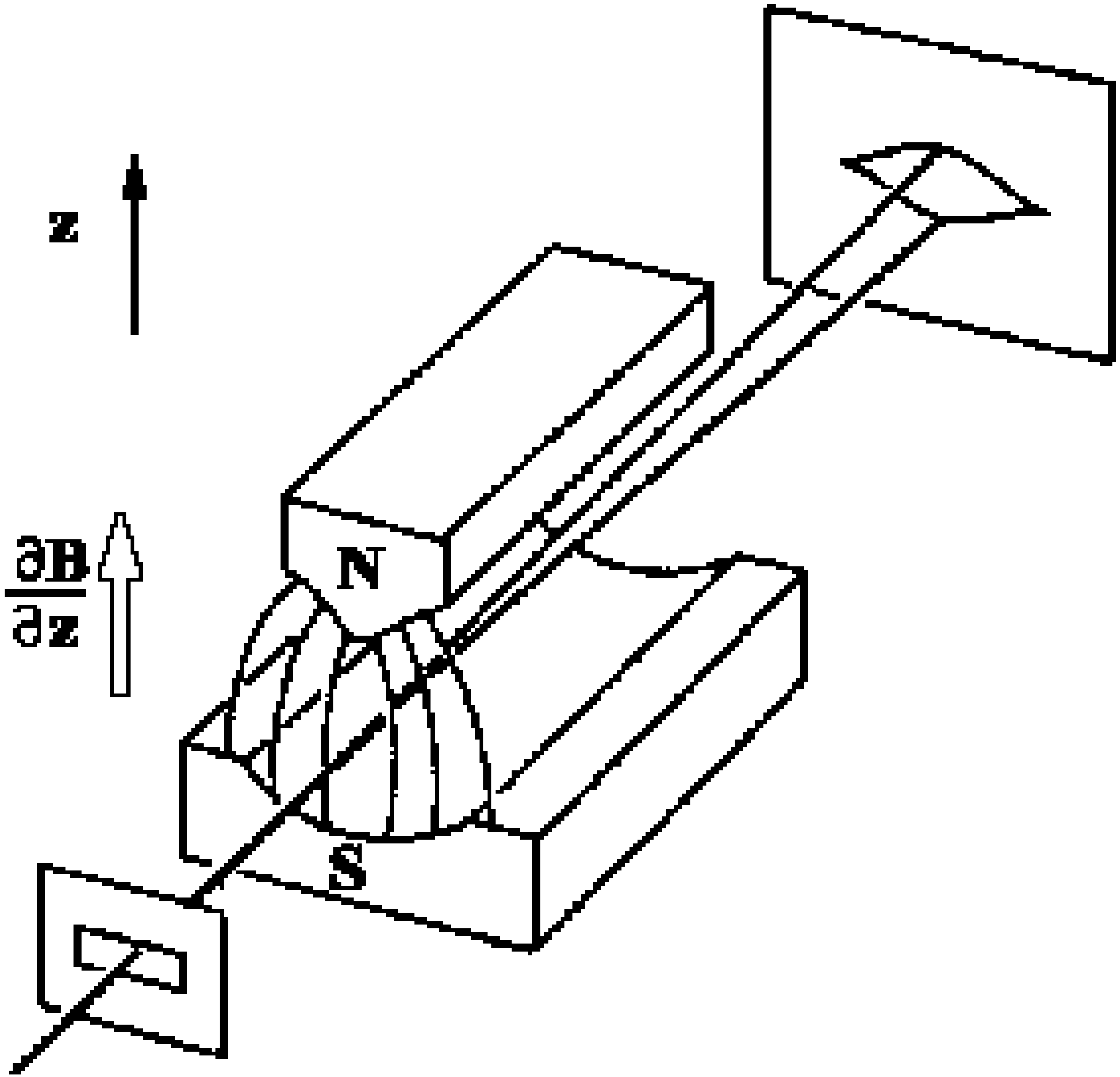
$$U = -\boldsymbol{\mu} \cdot \boldsymbol{B} + C \qquad \boldsymbol{F} = -\nabla U = \nabla(\boldsymbol{\mu} \cdot \boldsymbol{B})$$

$$\boldsymbol{B} \approx B_z(z)\boldsymbol{e}_z$$

$$\boldsymbol{F} = \nabla U \approx \nabla(\mu_z B_z) \approx \mu_z \frac{\partial B_z}{\partial z} \boldsymbol{e}_z$$

$$\sim L_z \sim m\hbar$$

Split $2\ell+1$


z
N
S
$\frac{\partial B}{\partial z}$

In a non-uniform magnetic field

$$\boldsymbol{F} = \mu \cdot \nabla \boldsymbol{B}$$

$$U = -\boldsymbol{\mu} \cdot \boldsymbol{B} + C \qquad \boldsymbol{F} = -\nabla U = \nabla(\boldsymbol{\mu} \cdot \boldsymbol{B})$$

$$\boldsymbol{B} \approx B_z(z)\boldsymbol{e}_z$$

$$\boldsymbol{F} = \nabla U \approx \nabla(\mu_z B_z) \approx \mu_z \frac{\partial B_z}{\partial z} \boldsymbol{e}_z$$

$$\sim L_z \sim m\hbar$$

Split $2\ell+1$

In 1921, O. Stern and W. Gerlach
to measure the magnetic moment of silver atoms.

Electronic configuration of Ag : $[\mathrm{Kr}](4\mathrm{d})^{10}5\mathrm{s}$

No orbital angular moment

They found the beam splits into two components.

$2\ell+1=2?$

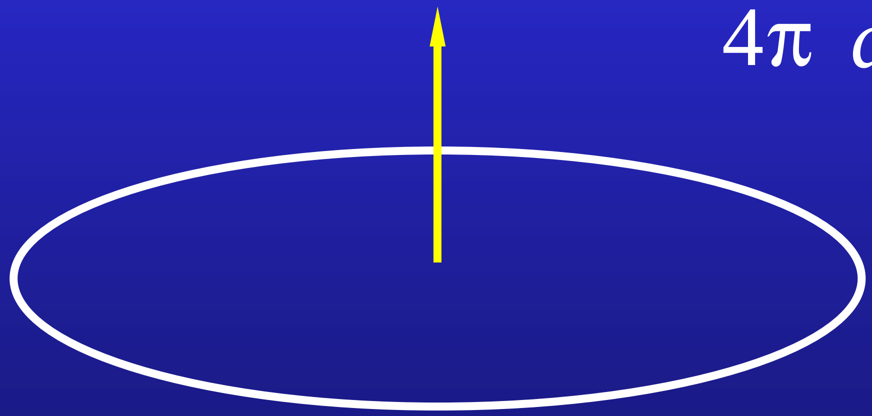
In 1926 Uhlenbeck and Goudsmit introduced the idea of electron spin angular momentum. For electron, the spin S may have values:

$$\left(\frac{\boldsymbol{S}}{\hbar}\right)_z=\pm\frac{1}{2} \qquad \textbf{and} \qquad \mu_{\mathrm{S}}=-\frac{e}{m}\boldsymbol{S}$$

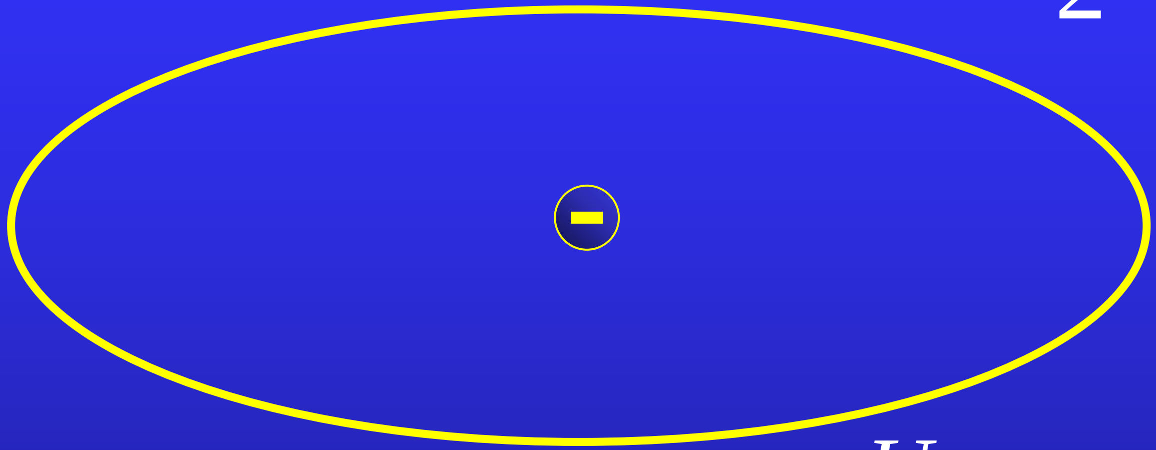
Spin is an intrinsic angular momentum. It is never associated with angular coordinates.

How the spin interprets the fine structure?

$$\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{2\mu}{a^3} = \frac{\mu_0}{2}\frac{I\boldsymbol{k}}{a}$$



$$B = \frac{\mu_0}{2}\frac{I}{r} = \frac{\mu_0}{2}\frac{1}{r}\frac{ev}{2\pi r}$$



$$U = -\boldsymbol{\mu}_s \cdot \boldsymbol{B}$$

$$= \pm\frac{1}{2}\frac{e\hbar}{m}\times\frac{\mu_0}{2}\frac{1}{r}\frac{ev}{2\pi r}$$

$$\Delta U = 2\mu B$$

$$\Delta U = \frac{e\hbar}{m} \times \frac{\mu_0}{2} \frac{1}{r} \frac{ev}{2\pi r}$$

$$= \frac{\mu_0}{4\pi} \frac{e^2\hbar}{m} \frac{v}{r^2} \frac{mr}{mr}$$

$$= \frac{\mu_0}{4\pi} \frac{e^2\hbar}{m^2} \frac{n\hbar}{n^6 a_0^3} \qquad a_0 = \frac{\hbar^2}{me_s^2}, \qquad \alpha = \frac{e_s^2}{\hbar c}$$

$$\Delta U = 2\mu B = mc^2 \alpha^4 \frac{1}{n^5}$$

For $n = 2$, it is 4.53×10^{-5}eV

The observed split of Lyman α is 4.54×10^{-5}eV

- $$K = mc^2 - m_0c^2 = m_0c^2\left(\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} - 1\right) \approx$$

- $\boldsymbol{E}' = \cdots$

  $\boldsymbol{B}' = \cdots$

L. H. Thomas

Nature, **117**(1926)

Electron never spins.

a uniform ball of charge e and of radius of classical electron radius r_0 when spins with angular velocity ω

$$\mu=\int\rho\,\mathrm{d}\tau\,\frac{\omega}{2\pi}\pi\left(r\sin\theta\right)^2$$

$$=\rho\frac{\omega}{2}\int_0^{r_0}r^4\mathrm{d}r\int_{-1}^{1}\sin^2\theta\,\mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\varphi$$

$$=\rho\frac{\omega}{2}\cdot\frac{1}{5}r_0^5\cdot\frac{4}{3}\cdot 2\pi=\frac{1}{5}\omega\, r_0\times e r_0$$

$$\rho=\frac{e}{\frac{4}{3}\pi r_0^3}$$

Let it equal to electron's spin magnetic moment:
(i.e. assume the spin magnetic moment is caused by such mechanism)

$$\frac{1}{5}\omega r_0 \times e r_0 = \frac{1}{2}\frac{e}{m}\hbar$$

$$\omega r_0 = \frac{5}{2}\frac{\hbar}{m r_0} = \frac{5}{2\alpha}c \qquad \alpha = \frac{1}{137}$$

It means that the surface speed is hundred times of light speed. Whereas for classical theory, substantial disaster is radiation due to acceleration rather than high speed.

25.5 Many-electron atom

Pauli's exclusion principle (1925):

• No two fermions (particles with half-integer spin) in a single system can exist in the same quantum state.

For electrons in an atom

No two electrons in a single atom can have the same set of quantum numbers (n, ℓ, m, m_s).

Why Pauli's principle?

Identical particles are indistinguishable.

If N particles are distinguishable, we can denote them by $\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_i, \cdots, \boldsymbol{r}_j, \cdots, \boldsymbol{r}_N$. Then

$$\left|\psi\left(\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_i, \cdots, \boldsymbol{r}_j, \cdots, \boldsymbol{r}_N\right)\right|^2 d^3 r_1 d^3 r_2 \cdots d^3 r_N$$

is the probability to find particle 1 in d^3r_1 in the vicinity of $\boldsymbol{r}_1$, particle 2 in d^3r_2 in the vicinity of $\boldsymbol{r}_2$, …and particle N in d^3r_N in the vicinity of $\boldsymbol{r}_N$.

If N particles are indistinguishable, then

$$\left|\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right)\right|^2 d^3r_1 d^3r_2\cdots d^3r_N$$

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Due to the indistinguishability of the identical particles, exchange any pair of the coordinates will not alter the probability density, i. e.

$$\left|\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right)\right|^2=\left|\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_N\right)\right|^2$$

Therefore, we have that

$$\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right) \text{ and } \psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_N\right)$$

represent the same quantum state.

Then, we have

$$\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right)=\lambda\,\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_N\right)$$
$$=\lambda^2\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right).$$

$$\lambda^2=1,\qquad \lambda=\pm 1.$$

Therefore, we have

$$\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right)=\mp\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_N\right)$$

$-$	antisymmetric	for fermions
$+$	symmetric	for bosons

For two independent identical particles (No interaction), the Hamiltonian can be written as

$$\hat{H}(\boldsymbol{r}_1, \boldsymbol{r}_2) = \hat{H}_0(\boldsymbol{r}_1) + \hat{H}_0(\boldsymbol{r}_2)$$

The single particle energy levels and eigenfunctions of the Hamiltonian are

$$E_1,\ E_2,\ \ E_3, \cdots E_m, \cdots;$$

$$\psi_1,\ \psi_2,\ \ \psi_3, \cdots \psi_m, \cdots$$

The energy levels and eigenfunctions satisfy the single-particle stationary Schrodinger equation

$$\hat{H}_0(\boldsymbol{r})\psi_m = E_m \psi_m .$$

One can easily see that

$$\hat{H}(\boldsymbol{r}_1,\boldsymbol{r}_2)\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)=\left(\hat{H}_0(\boldsymbol{r}_1)+\hat{H}_0(\boldsymbol{r}_2)\right)\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)$$

$$=\psi_n(\boldsymbol{r}_2)\hat{H}_0(\boldsymbol{r}_1)\psi_m(\boldsymbol{r}_1)+\psi_m(\boldsymbol{r}_1)\hat{H}_0(\boldsymbol{r}_2)\psi_n(\boldsymbol{r}_2)$$

$$=\left(E_m+E_n\right)\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)$$

$\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)$ is an eigenfunction of $\hat{H}(\boldsymbol{r}_1,\boldsymbol{r}_2)$.

Is $\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)$ the wavefunction of the two particle system? Assume $m \neq n$.

Noting that

$$\hat{H}(\boldsymbol{r}_1,\boldsymbol{r}_2)\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)=\left(E_m+E_n\right)\psi_m(\boldsymbol{r}_2)\psi_n(\boldsymbol{r}_1)$$

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we have

$$\psi_{S,A}\left(\boldsymbol{r}_1,\boldsymbol{r}_2\right)=\frac{1}{\sqrt{2}}\left(\psi_m(\boldsymbol{r}_1)\psi_n(\boldsymbol{r}_2)\pm\psi_m(\boldsymbol{r}_2)\psi_n(\boldsymbol{r}_1)\right)$$

Symmetric bosons

Antisymmetric fermions

$$\psi_A(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_m(\boldsymbol{r}_1) & \psi_m(\boldsymbol{r}_2) \\ \psi_n(\boldsymbol{r}_1) & \psi_n(\boldsymbol{r}_2) \end{vmatrix}$$

Slater determinant

If the single particle wave functions for N independent identical fermions are

$$\psi_a(\boldsymbol{r}), \psi_b(\boldsymbol{r}), \psi_c(\boldsymbol{r}), \cdots$$

the antisymmetric)N particle wave function is

$$\psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_a(\boldsymbol{r}_1) & \psi_a(\boldsymbol{r}_2) & \cdots & \psi_a(\boldsymbol{r}_N) \\ \psi_b(\boldsymbol{r}_1) & \psi_b(\boldsymbol{r}_2) & \cdots & \psi_b(\boldsymbol{r}_N) \\ \psi_c(\boldsymbol{r}_1) & \psi_c(\boldsymbol{r}_2) & \cdots & \psi_c(\boldsymbol{r}_N) \\ \cdots & \cdots & & \end{vmatrix}$$

Slater determinant

$$\psi(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_N)=\frac{1}{\sqrt{N!}}\begin{vmatrix}\psi_a(\boldsymbol{r}_1) & \psi_a(\boldsymbol{r}_2) & \cdots & \psi_a(\boldsymbol{r}_N)\\ \psi_b(\boldsymbol{r}_1) & \psi_b(\boldsymbol{r}_2) & \cdots & \psi_b(\boldsymbol{r}_N)\\ \psi_c(\boldsymbol{r}_1) & \psi_c(\boldsymbol{r}_2) & \cdots & \psi_c(\boldsymbol{r}_N)\\ \cdots & \cdots & & \end{vmatrix}$$

Slater determinant

b=c or $\ldots\Rightarrow$ $\psi(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_N)\equiv 0.$

$\Downarrow$

Pauli's principle!

Shell model

quantum numbers $n \quad l \quad m \quad m_s$

The states with n=1,2,3,4,… construct atomic shells denoted as K, L, M, N, ….

The States with a certain value of n and ℓ are known as subshells.

$n\ell \qquad \ell$=0,1,2,3,4,… $\longleftrightarrow$ $s,p,d,f,g, \ldots$

two rules for filling subshells:

- The capacity of each subshell is $2(2\ell + 1)$;
- The electrons will occupy the lowest energy states available.

n	1	2	2	3	3	4	3	4	5	4	…
ℓ	0	0	1	0	1	0	2	1	0	2	…
subshell	1s	2s	2p	3s	3p	4s	3d	4p	5s	4d	…
$2(2\ell+1)$	2	2	6	2	6	2	10	6	2	10	…

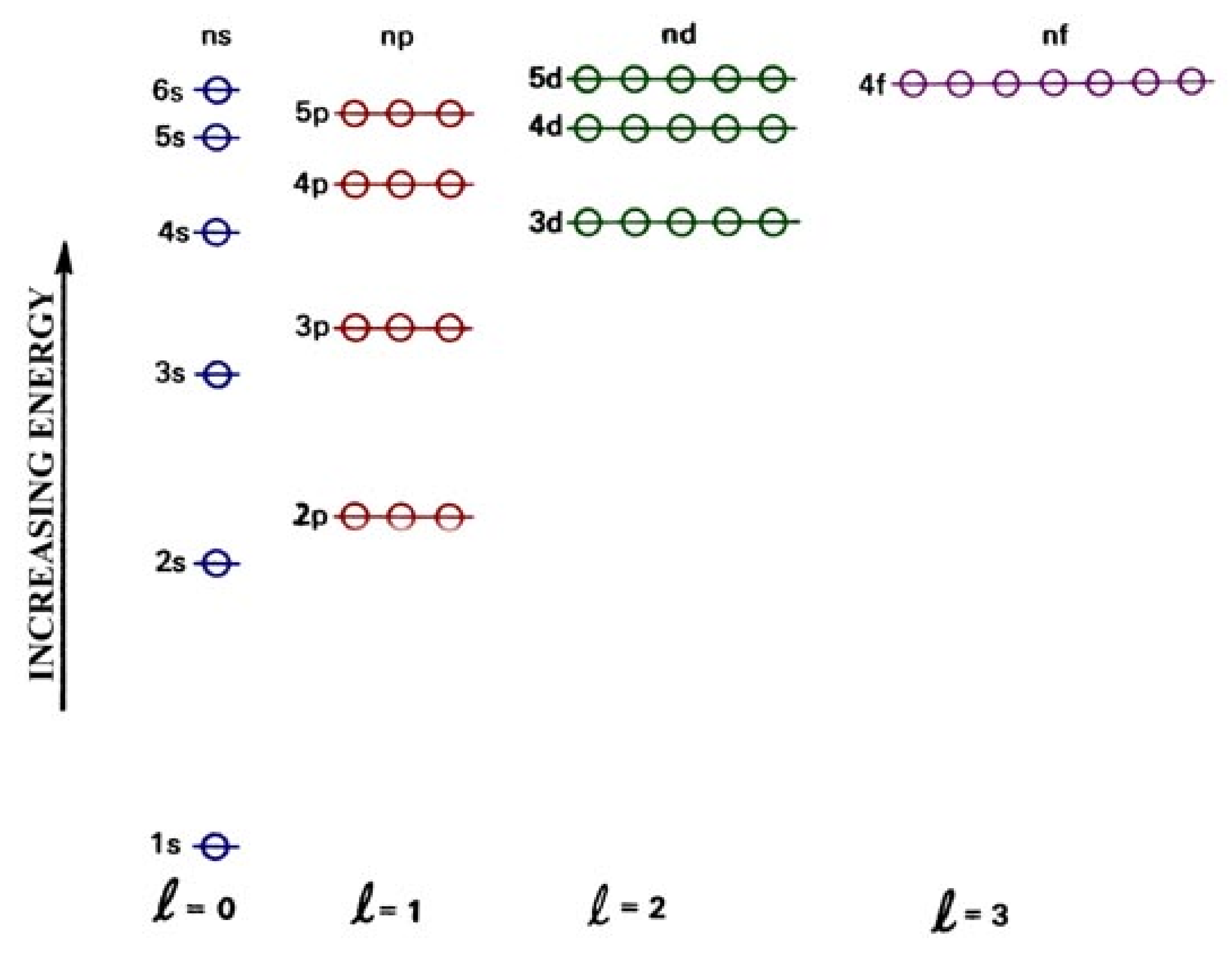
Hydrogen-like atoms $\longrightarrow$ many-electron atoms

Coulomb potentials $\longrightarrow$ Effective core potentials

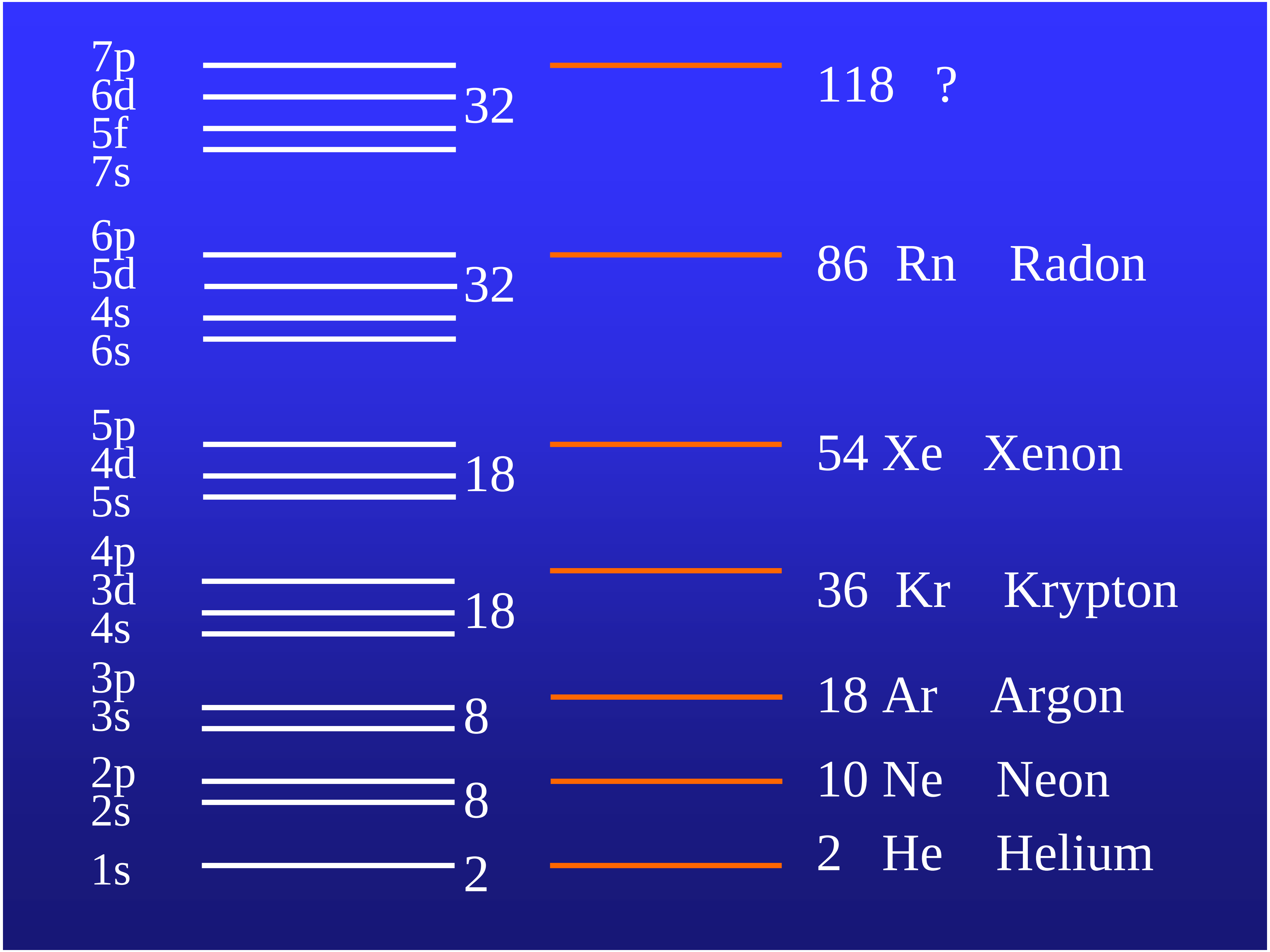
(atomic core)

$$E_n \longrightarrow E_{nl}$$

Shell Structure of Atomic Energy Levels



ns
np
nd
nf
6s
5s
4s
3s
2s
1s
5p
4p
3p
2p
5d
4d
3d
4f
INCREASING ENERGY
$\ell = 0$
$\ell = 1$
$\ell = 2$
$\ell = 3$


7p
6d
5f
7s
32
118 ?
6p
5d
4s
6s
32
86 Rn Radon
5p
4d
5s
18
54 Xe Xenon
4p
3d
4s
18
36 Kr Krypton
3p
3s
8
18 Ar Argon
2p
2s
8
10 Ne Neon
1s
2
2 He Helium

Why $E_{4s} < E_{3d}$? $E_{nl} = -13.6\, Z_{eff}^2 / n^2$

4s electron has the highly penetrating orbit. $Z_{eff} \uparrow$

The groundstate electronic configurations

1 H $1s$

2 He $1s^2$

3 Li $1s^22s$ $[He]2s$

…

10 Ne $[He]2s^22p^6$

11 Na $[Ne]2s$

…

18 Ar $[Ne]\ 3s^23p^6$

19 K $[Ar]\ 4s$

20 Ca $[Ar]\ 4s^2$

scandium 钪

21 Sc $[Ar]\ 4s^23d$

3d level is slightly lower than 4s.

…

29 Cu $[Ar]3d^{10}4s$

30 Zn $[Ar]\ 4s^23d^{10}$

…

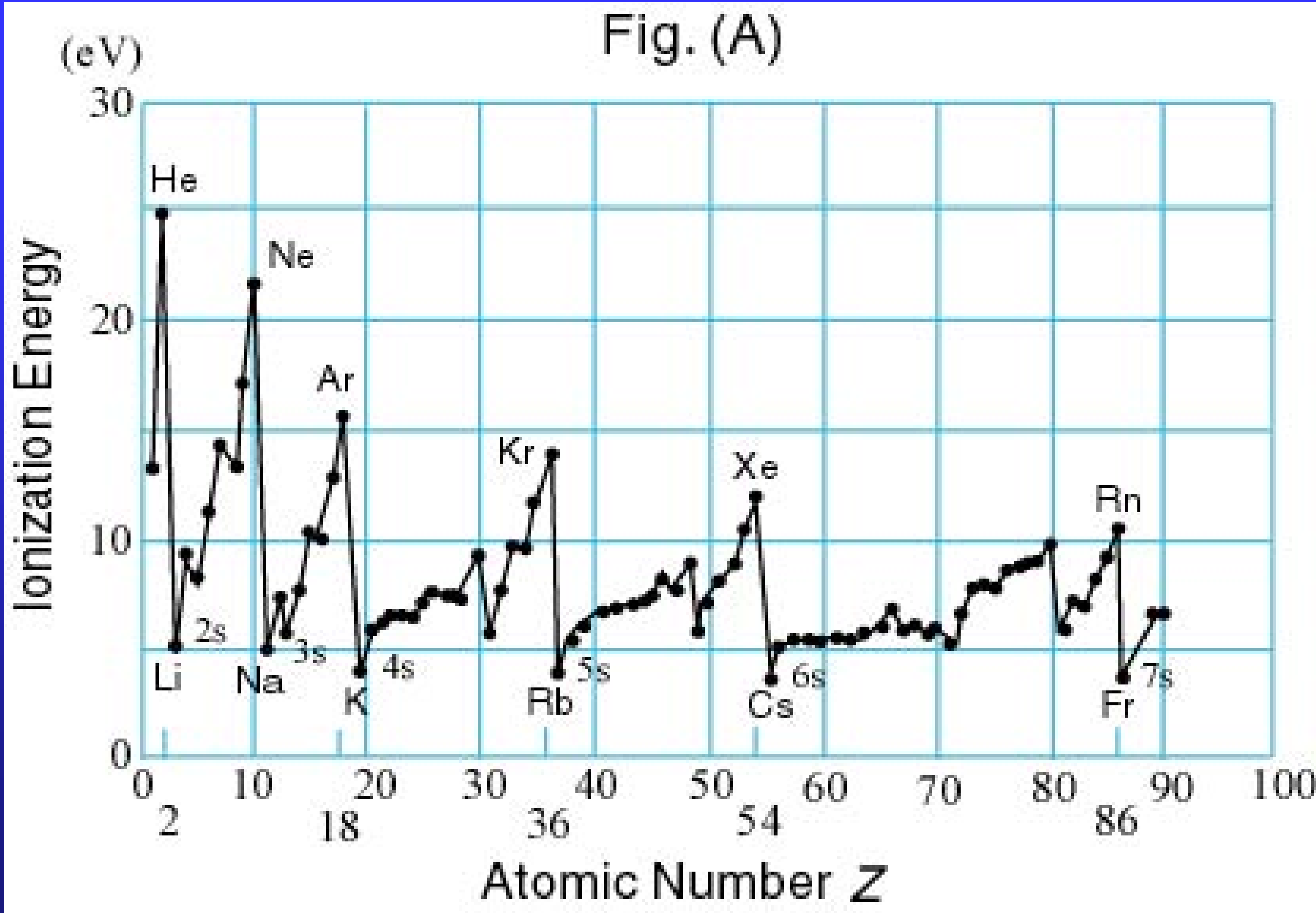
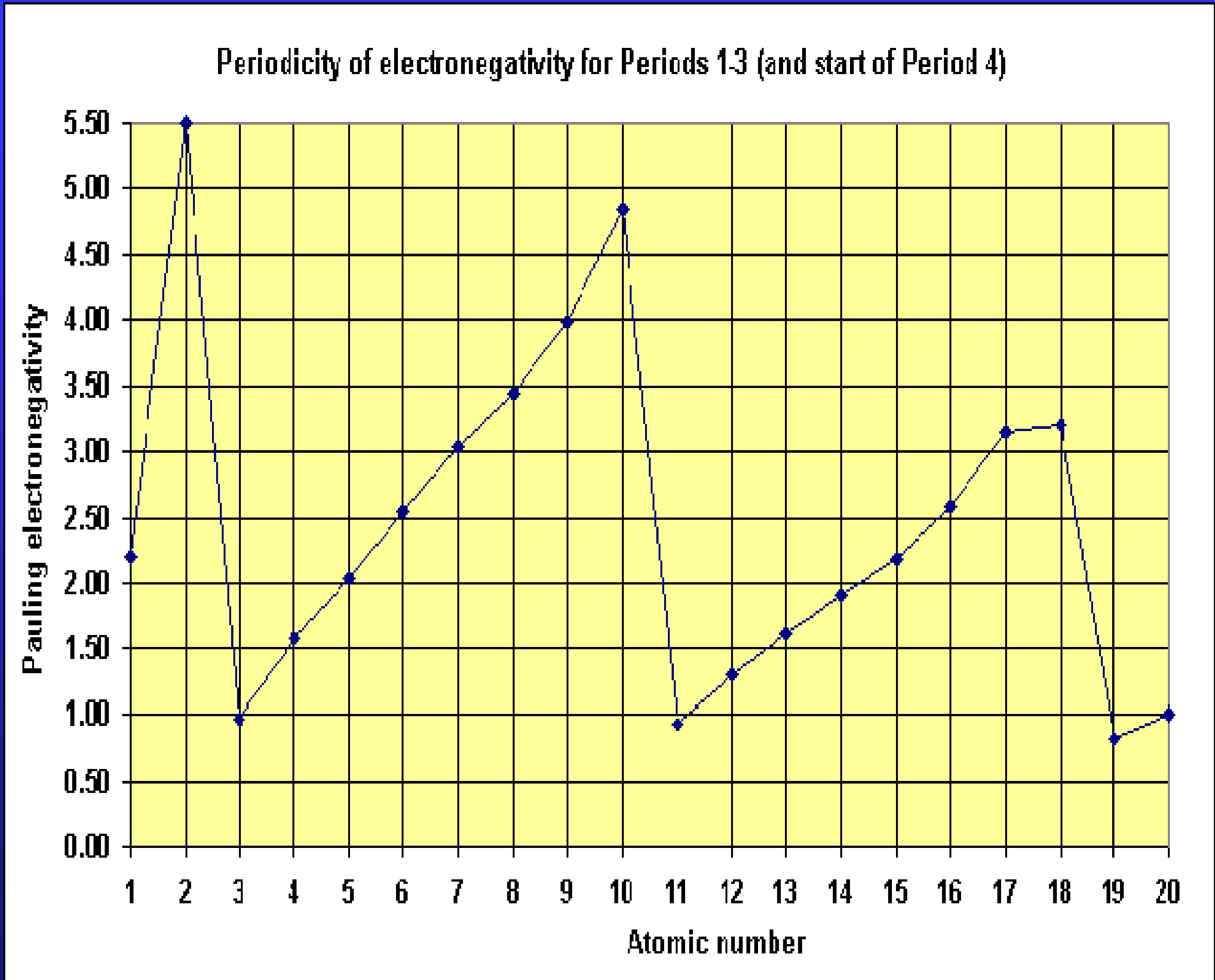
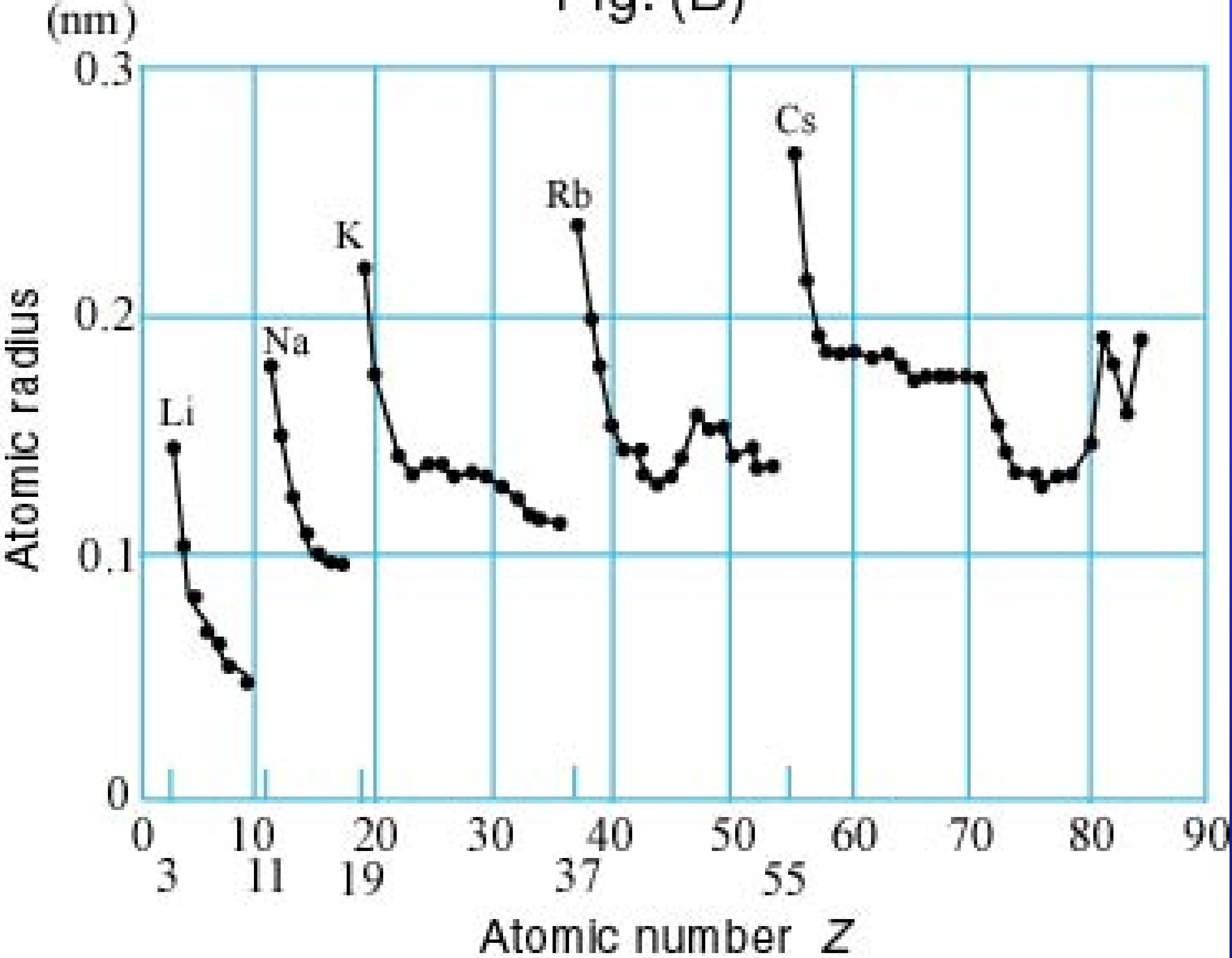


Fig. (A)
(eV)
Ionization Energy
30
20
10
0
He
Ne
Ar
Kr
Xe
Rn
Li
Na
K
Rb
Cs
Fr
2s
3s
4s
5s
6s
7s
0
10
20
30
40
50
60
70
80
90
100
2
18
36
54
86
Atomic Number Z



Periodicity of electronegativity for Periods 1-3 (and start of Period 4)
Pauling electronegativity
5.50
5.00
4.50
4.00
3.50
3.00
2.50
2.00
1.50
1.00
0.50
0.00
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Atomic number

Fig. (B)

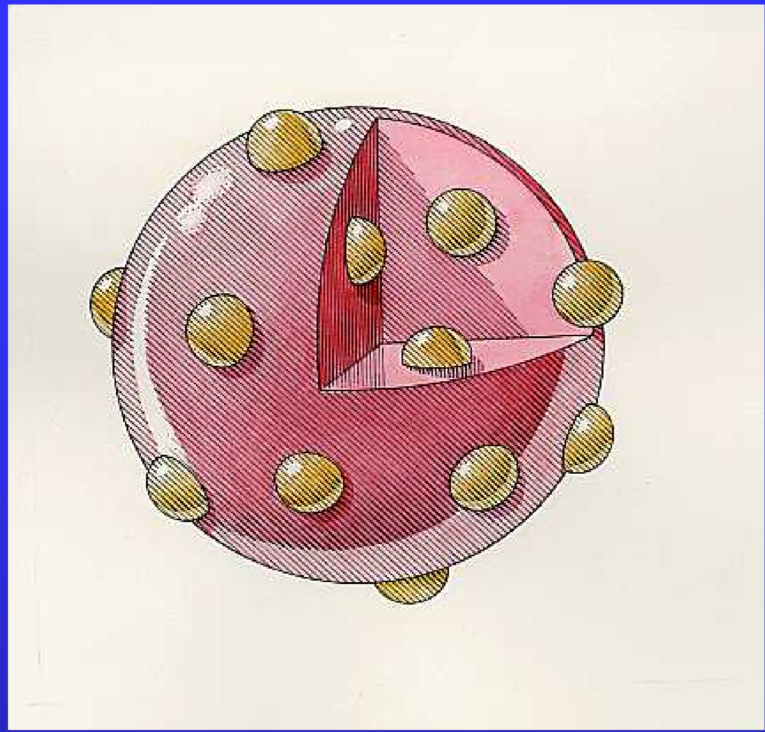


(nm)
0.3
0.2
0.1
0
Atomic radius
Li
Na
K
Rb
Cs
0
10
20
30
40
50
60
70
80
90
3
11
19
37
55
Atomic number Z

In Periodic table, the elements were organized into groups and periods based on their physical and chemical properties by Mendeleev in 1859. It is a great triumph for atomic theory to understand that organization in terms of atomic levels (or quantum states).

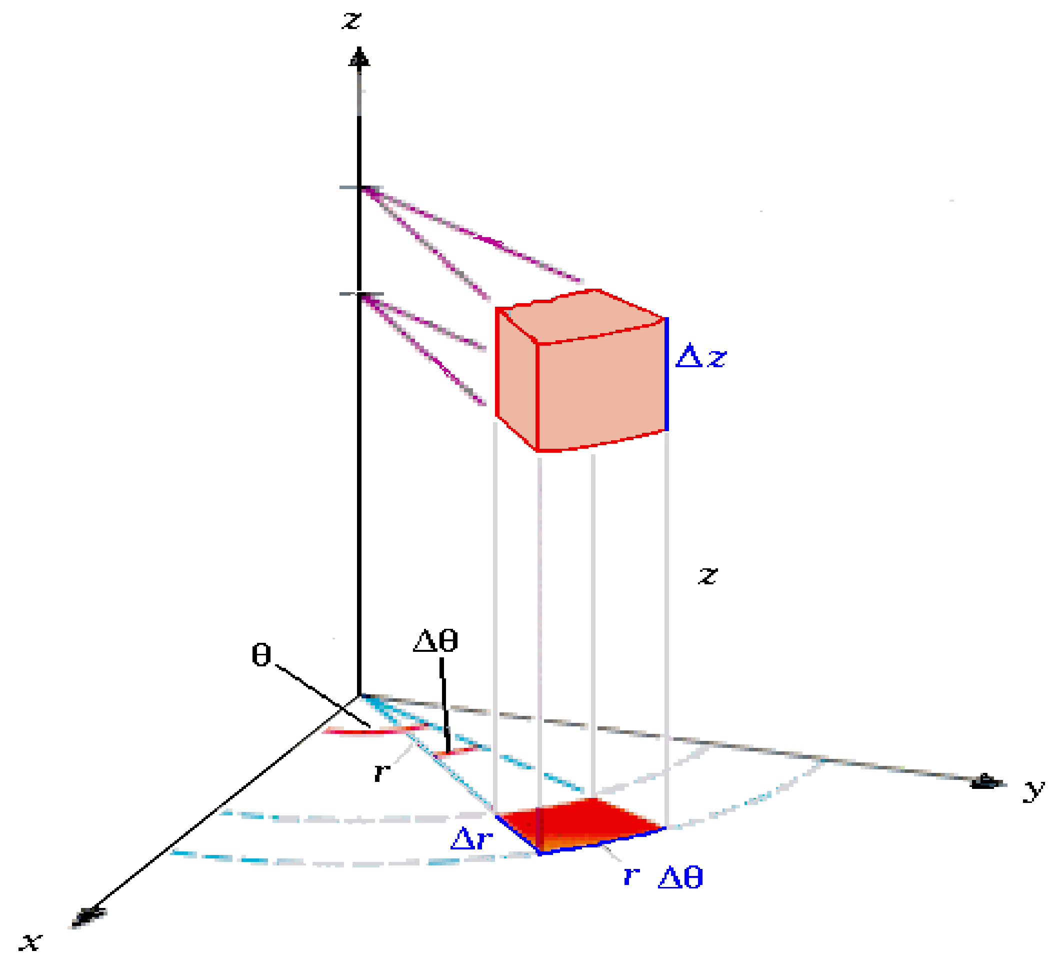
Assignment: 25.6, 25.7

A traditional British plum pudding



J.J. Thomson's
'Plum Pudding Model'
of the atom

Cylindrical coordinates



$$\Delta V = (r\Delta\theta)(\Delta r)(\Delta z)$$

$$\Rightarrow dV = r dr\, d\theta\, dz$$

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For electrons in an atom

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If N particles are distinguishable, we can denote them by $\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_i, \cdots, \boldsymbol{r}_j, \cdots, \boldsymbol{r}_N$. Then

$$\left|\psi\left(\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_i, \cdots, \boldsymbol{r}_j, \cdots, \boldsymbol{r}_N\right)\right|^2 d^3 r_1 d^3 r_2 \cdots d^3 r_N$$

is the probability to find particle 1 in $d^3 r_1$ in the vicinity of $\boldsymbol{r}_1$, particle 2 in $d^3 r_2$ in the vicinity of $\boldsymbol{r}_2$, …and particle N in $d^3 r_N$ in the vicinity of $\boldsymbol{r}_N$.

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represent the same quantum state.

Then, we have

$$\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_N\right)=\lambda\,\psi\left(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_j,\cdots,\boldsymbol{r}_i,\cdots,\boldsymbol{r}_N\right)$$
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Slater determinant

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the antisymmetric *N* particle wave function is

$$\psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_a(\boldsymbol{r}_1) & \psi_a(\boldsymbol{r}_2) & \cdots & \psi_a(\boldsymbol{r}_N) \\ \psi_b(\boldsymbol{r}_1) & \psi_b(\boldsymbol{r}_2) & \cdots & \psi_b(\boldsymbol{r}_N) \\ \psi_c(\boldsymbol{r}_1) & \psi_c(\boldsymbol{r}_2) & \cdots & \psi_c(\boldsymbol{r}_N) \\ \cdots & \cdots & & \end{vmatrix}$$

Slater determinant

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Slater determinant

b=c or …$\Rightarrow$ $\psi(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots,\boldsymbol{r}_N)\equiv 0.$

$$\Downarrow$$

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quantum numbers $n \quad l \quad m \quad m_s$

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ℓ	0	0	1	0	1	0	2	1	0	2	…
subshell	1s	2s	2p	3s	3p	4s	3d	4p	5s	4d	…
$2(2\ell+1)$	2	2	6	2	6	2	10	6	2	10	…

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(atomic core)

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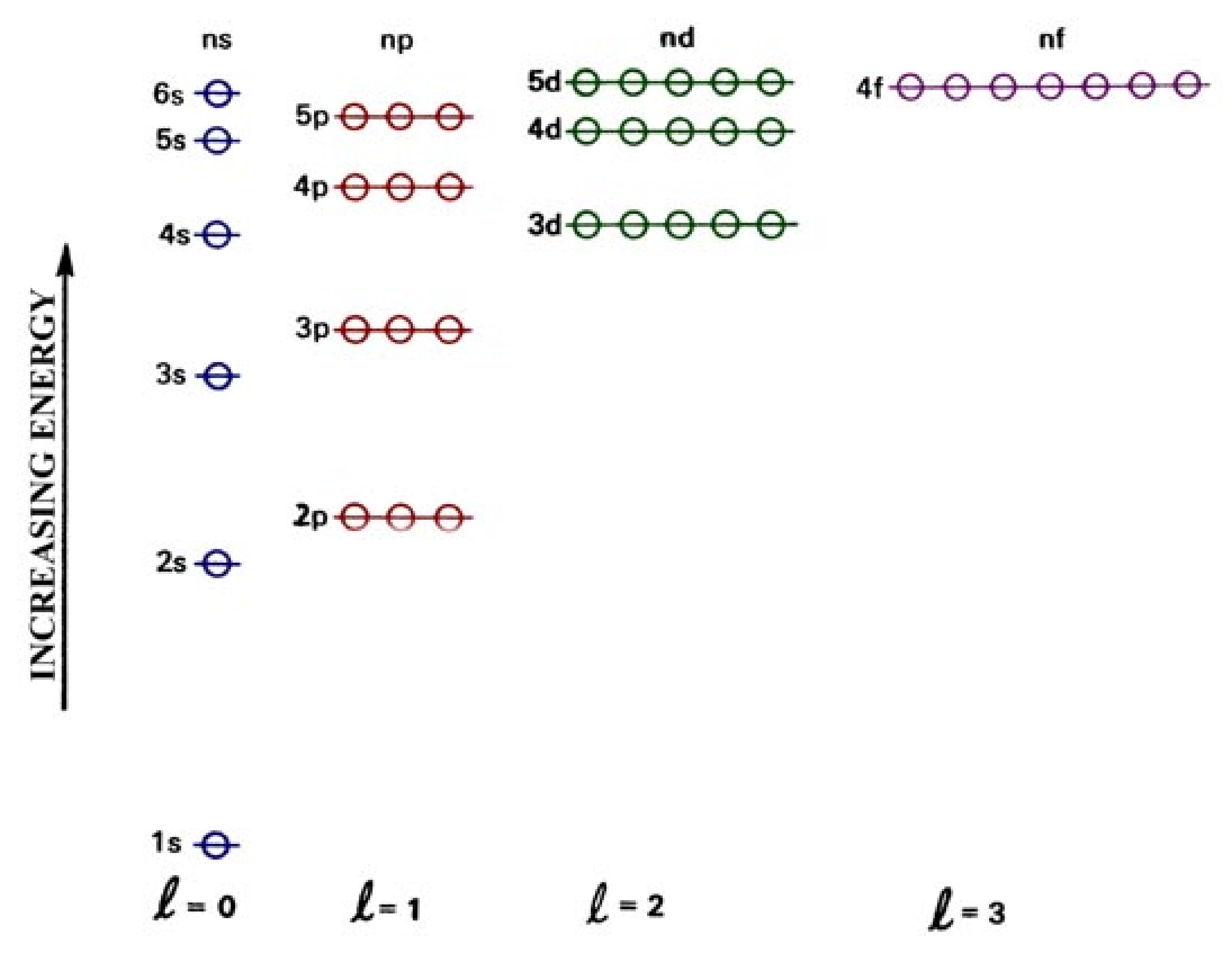
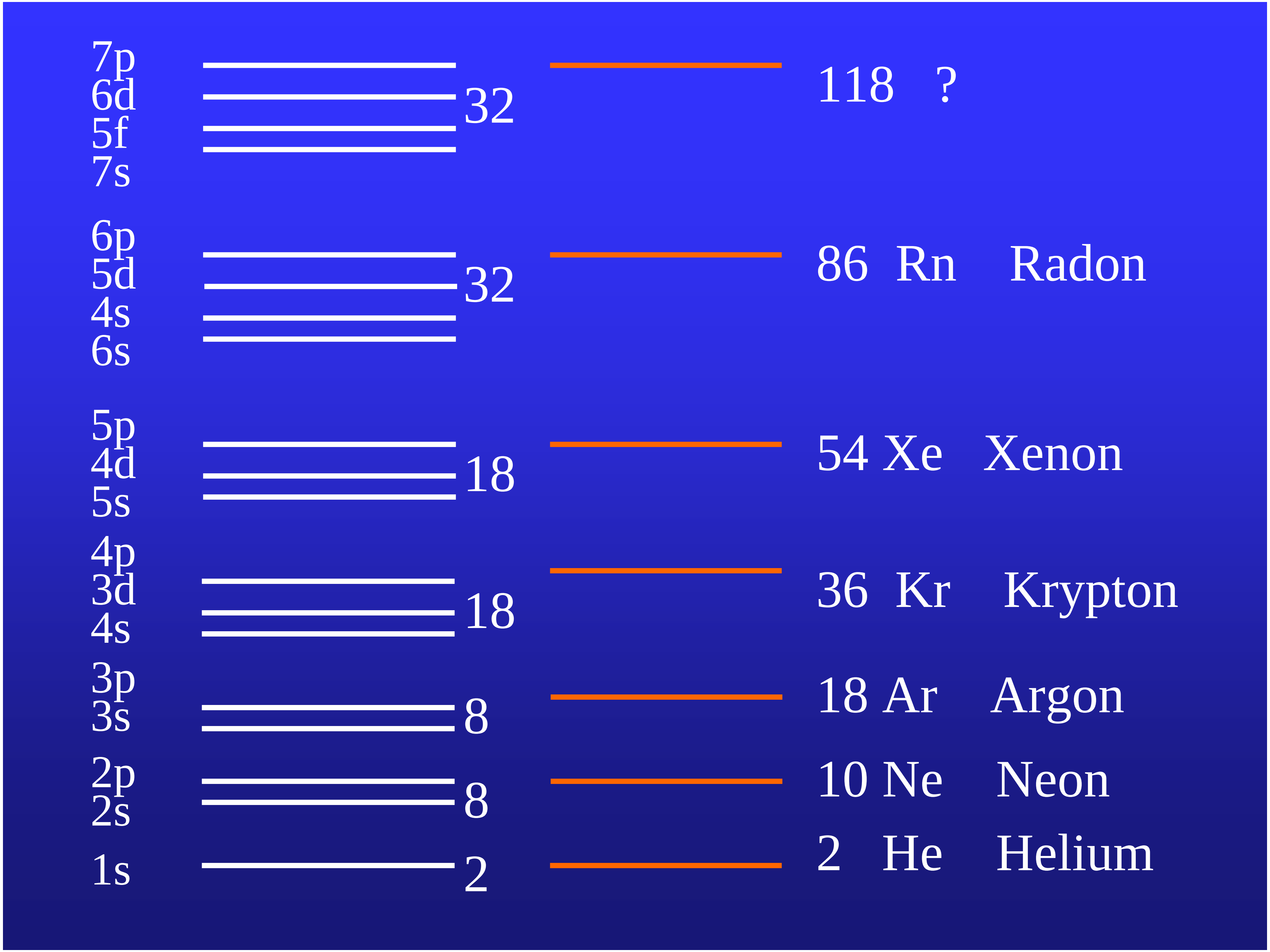
Shell Structure of Atomic Energy Levels



ns
np
nd
nf
6s
5s
4s
3s
2s
1s
5p
4p
3p
2p
5d
4d
3d
4f
INCREASING ENERGY
ℓ = 0
ℓ = 1
ℓ = 2
ℓ = 3


7p
6d
5f
7s
32
118 ?
6p
5d
4s
6s
32
86 Rn Radon
5p
4d
5s
18
54 Xe Xenon
4p
3d
4s
18
36 Kr Krypton
3p
3s
8
18 Ar Argon
2p
2s
8
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1s
2
2 He Helium

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…

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…

18 Ar [Ne] $3s^2 3p^6$

19 K [Ar] 4s

20 Ca [Ar] $4s^2$

scandium 钪

21 Sc [Ar] $4s^2 3d$

3d level is slightly lower than 4s.

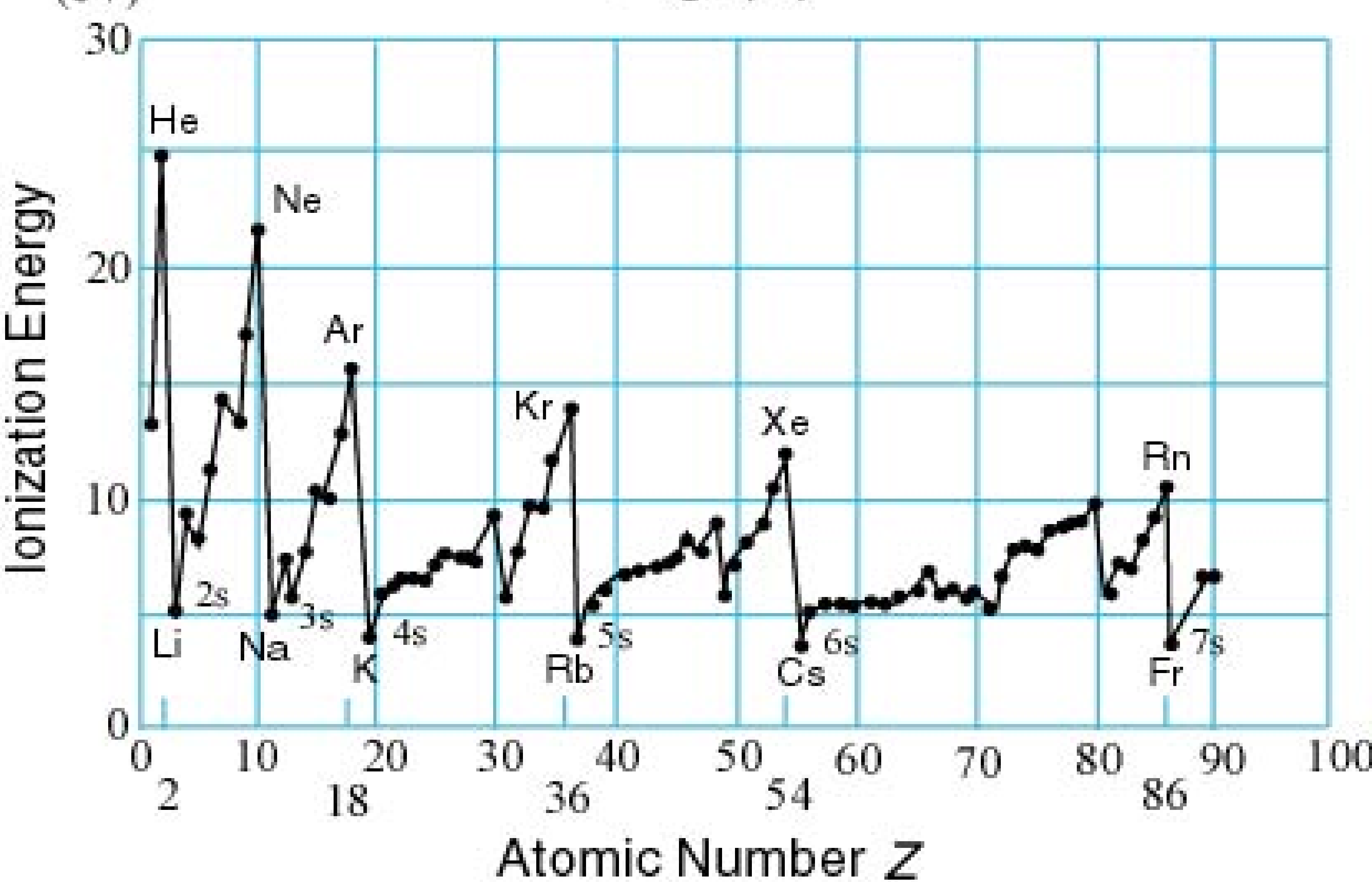
…

29 Cu [Ar]$3d^{10}4s$

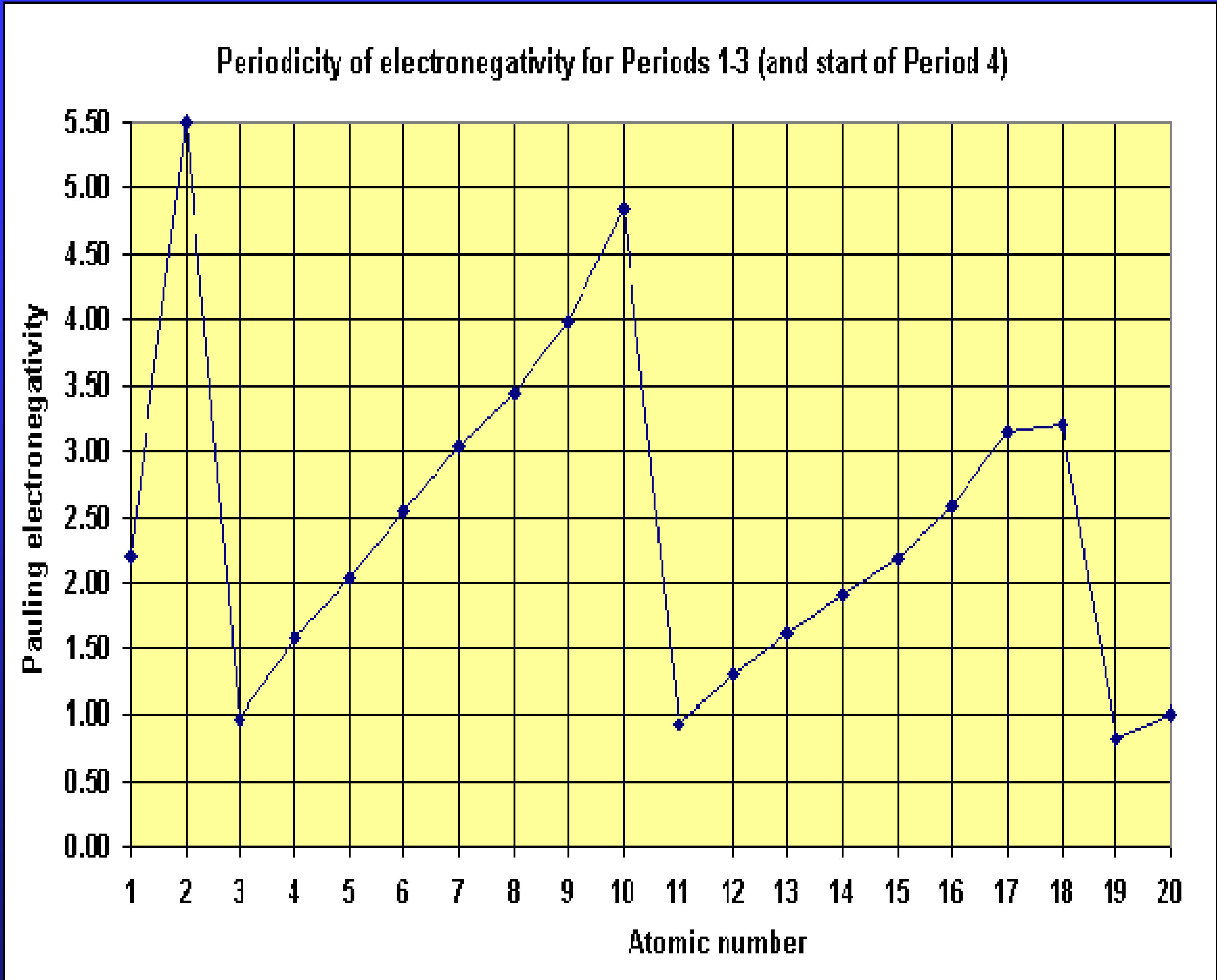
30 Zn [Ar] $4s^2 3d^{10}$

…

Fig. (A)



(eV)
Ionization Energy
30
20
10
0
He
Ne
Ar
Kr
Xe
Rn
Li
Na
K
Rb
Cs
Fr
2s
3s
4s
5s
6s
7s
0
10
20
30
40
50
60
70
80
90
100
2
18
36
54
86
Atomic Number Z


Periodicity of electronegativity for Periods 1-3 (and start of Period 4)
Pauling electronegativity
5.50
5.00
4.50
4.00
3.50
3.00
2.50
2.00
1.50
1.00
0.50
0.00
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Atomic number

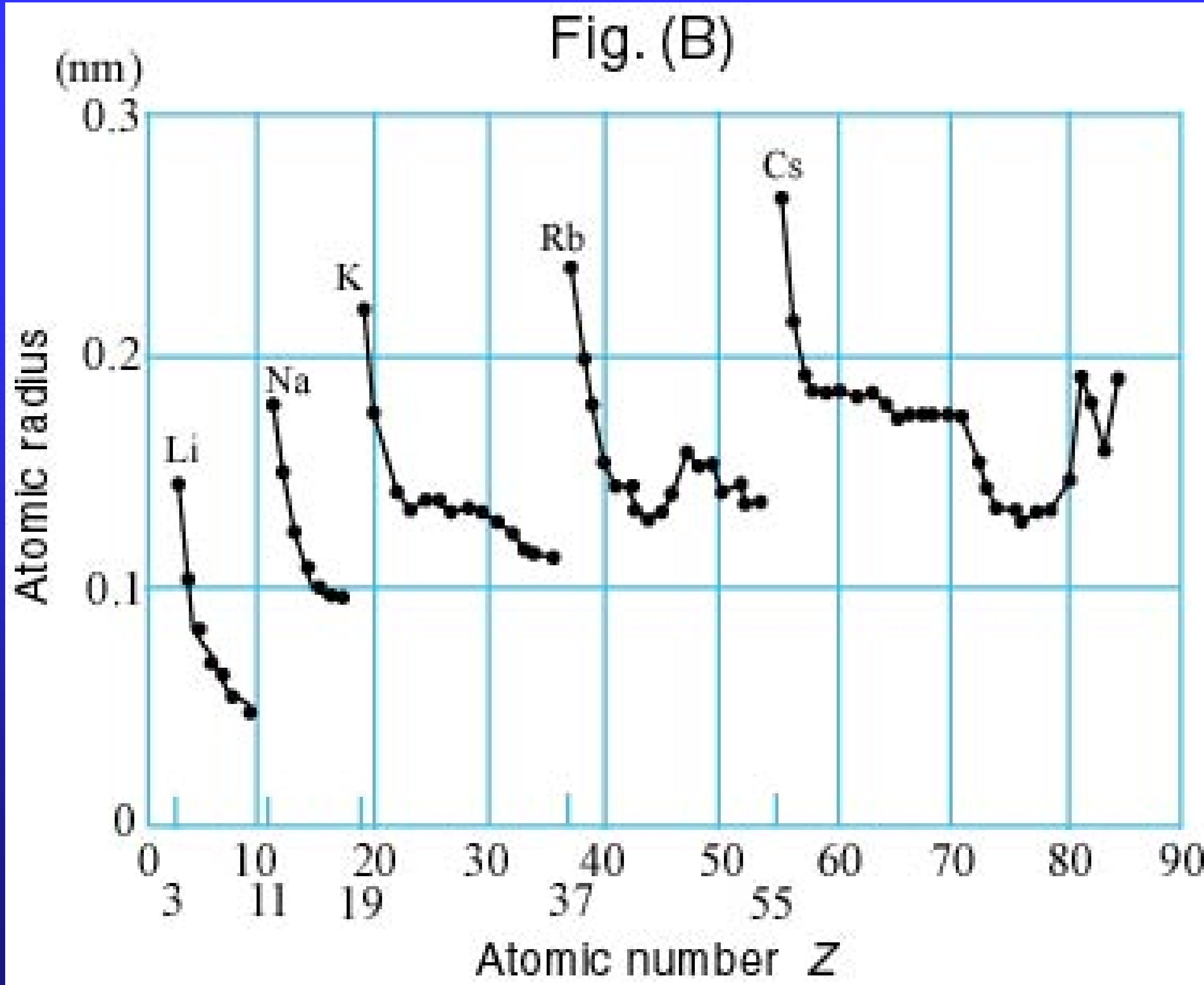
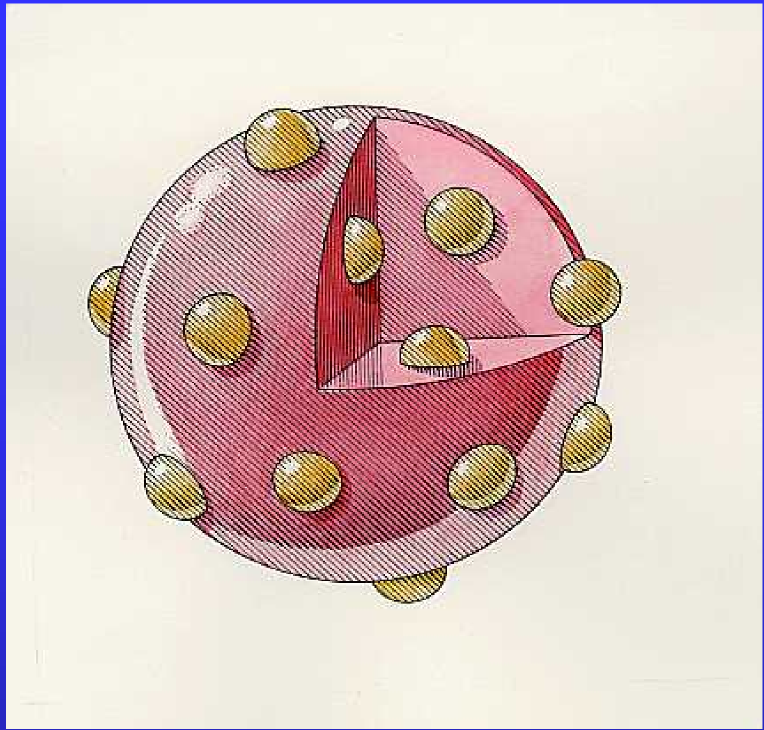


Fig. (B)
(nm)
0.3
0.2
0.1
0
Atomic radius
Li
Na
K
Rb
Cs
0 10 20 30 40 50 60 70 80 90
3 11 19 37 55
Atomic number Z

In Periodic table, the elements were organized into groups and periods based on their physical and chemical properties by Mendeleev in 1859. It is a great triumph for atomic theory to understand that organization in terms of atomic levels (or quantum states).

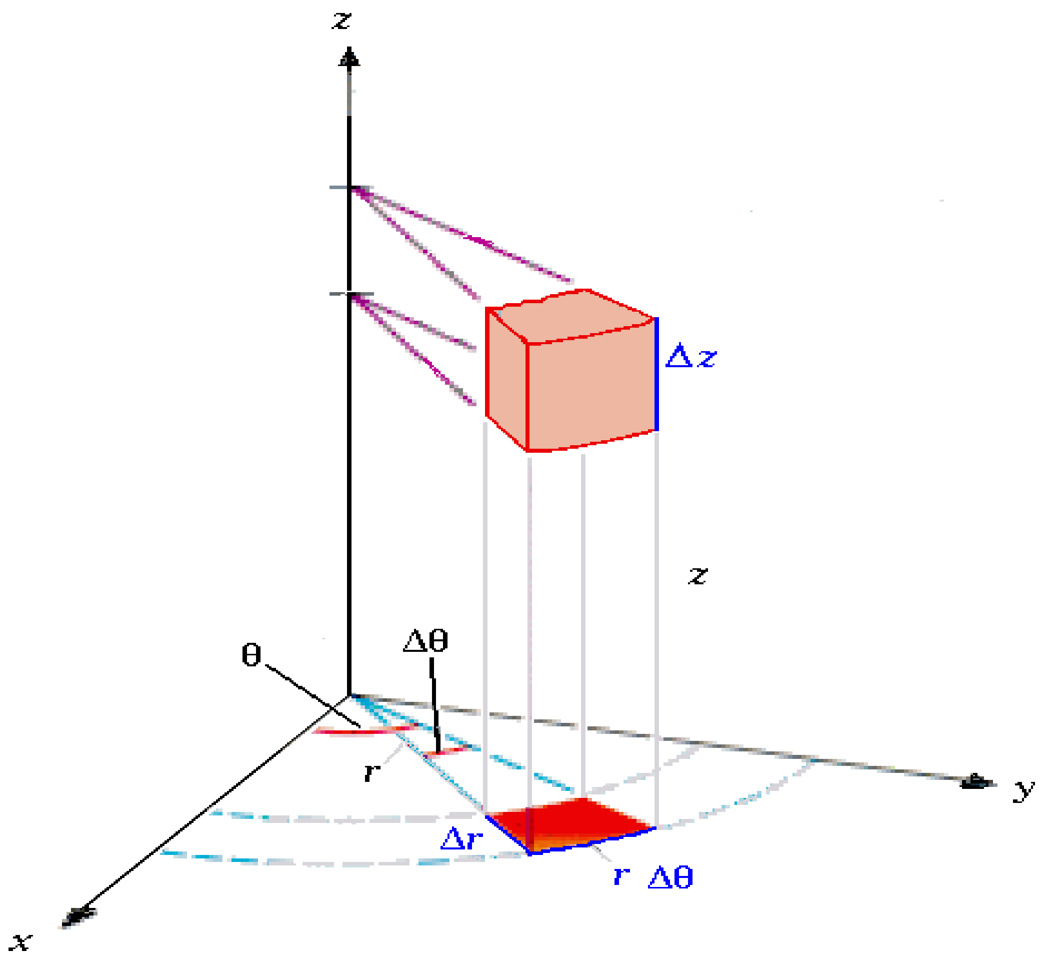
Assignment: 25.6, 25.7



A traditional British plum pudding

J.J. Thomson's **'Plum Pudding Model'** of the atom

Cylindrical coordinates



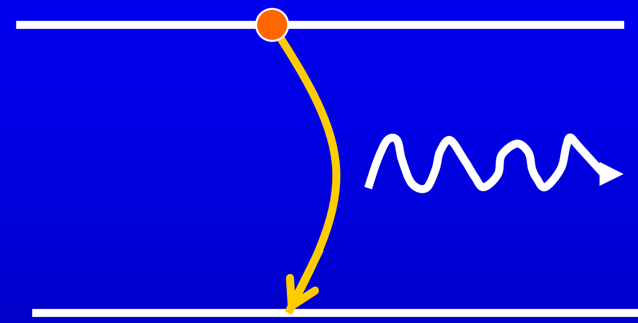
$$\Delta V = (r\Delta\theta)(\Delta r)(\Delta z)$$

$$\Rightarrow dV = r dr\, d\theta\, dz$$

25.6 Laser

Light Amplification by the Stimulated Emission of Radiation

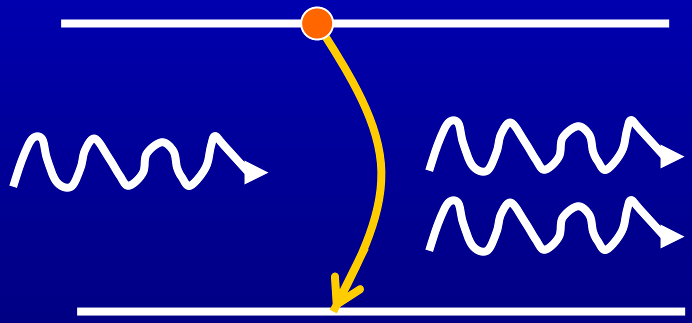
spontaneous emission



incoherent

stimulated emission

(induced)



coherent

$$h\nu = E_a - E_b$$ & Selection rules

According to Boltzmann statistics

$$E_2 > E_1$$

$$n_2 : n_1 = \exp\left[-(E_2 - E_1)/(k_B T)\right] < 1$$

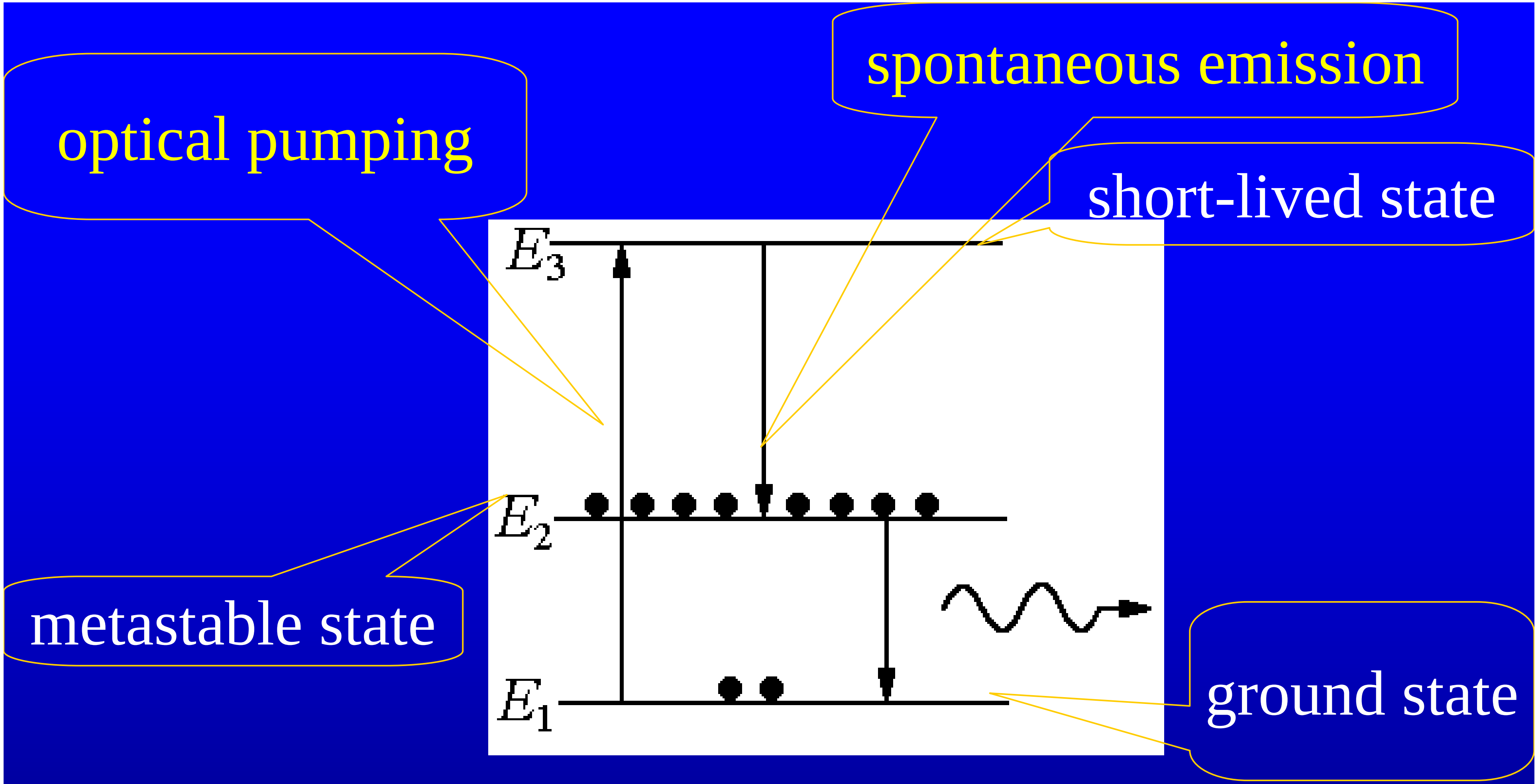
population inversion?

$$n_2 : n_1 = \exp\left[-(E_2 - E_1)/(k_B T)\right] > 1$$

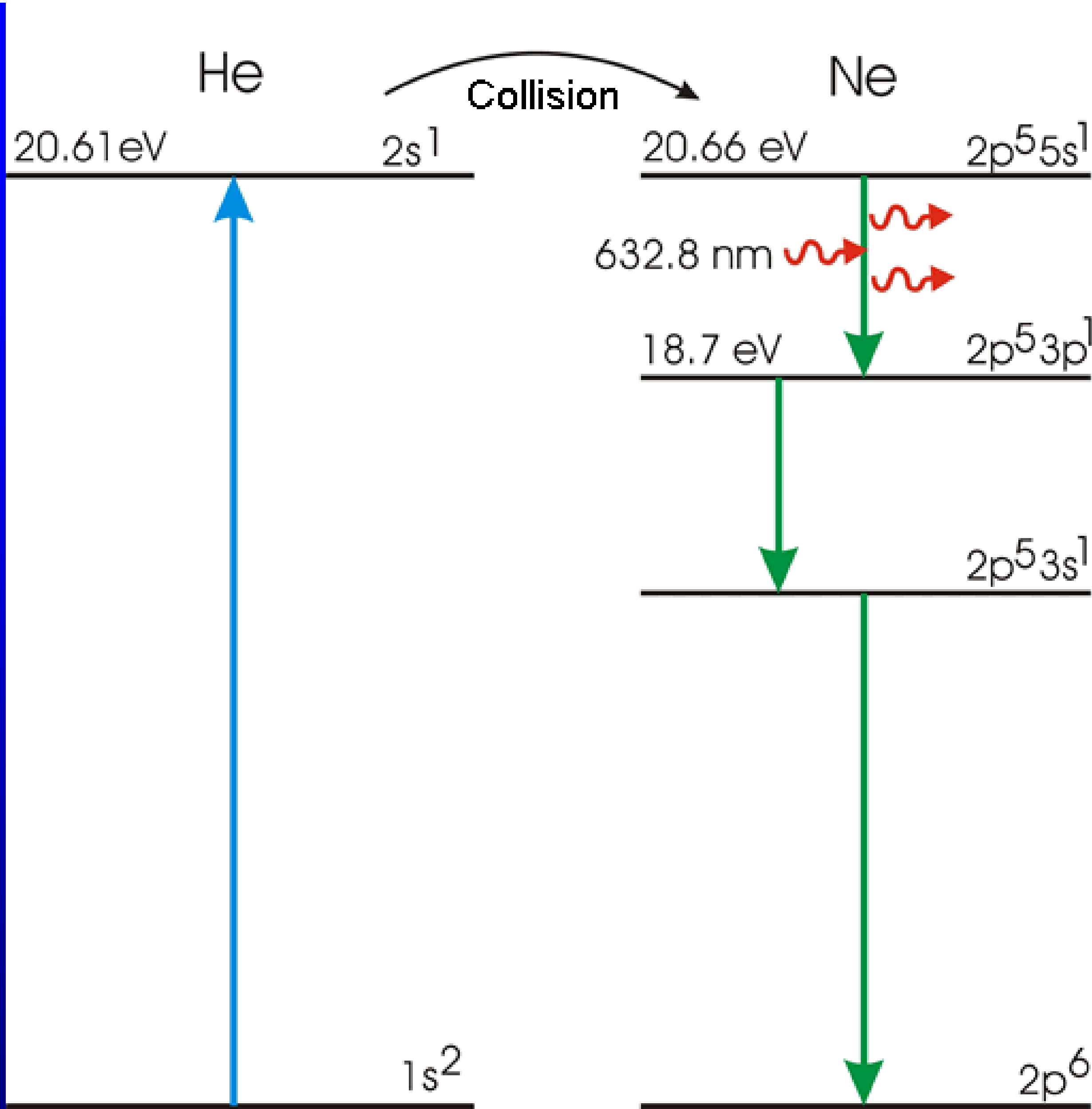
<u>minus temperature</u>

schemes for population inversion:

- optical pumping,
- collision of atoms with other stimulated atoms,
- collision of atom with electron and ion flows,
- transition of atoms from even higher excited state through spontaneous emission, or
- combination of some methods above.



Theodore Maiman invented first working LASER based on Ruby. May 16th 1960, Hughes Research Laboratories.



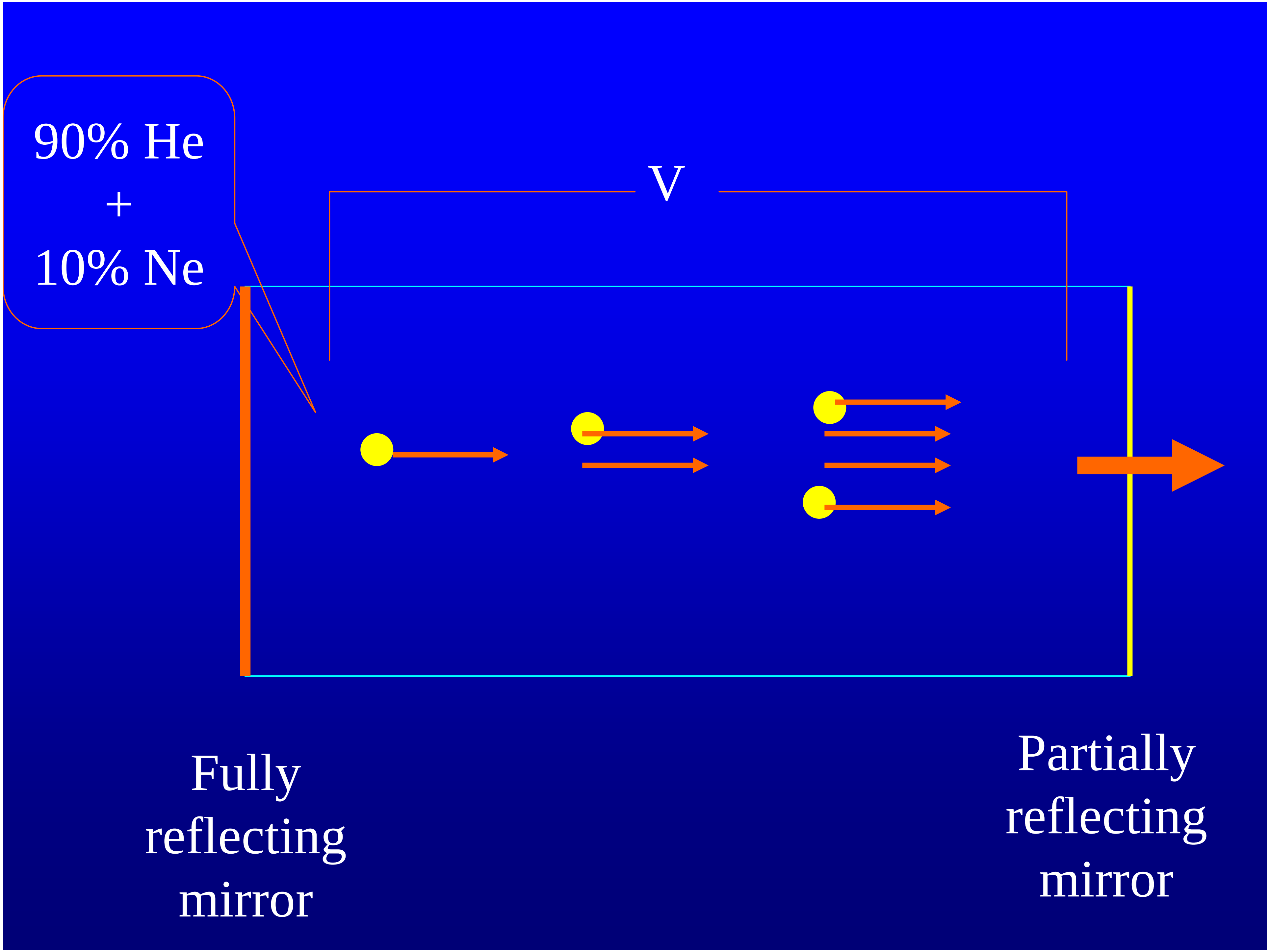
Helium Neon (HeNe) gas laser

$e + He = He^{*} + e$

an electron looses energy to He atom.

$He^{*} + Ne = Ne^{*} + He$

excited He collides with a Ne atom.


90% He
+
10% Ne
V
Fully reflecting mirror
Partially reflecting mirror

Ali Javan, William Bennet Jr. and Donald Herriot invented Helium Neon (HeNe) LASER at Bell Labs.

- Doped insulator laser
- Semiconductor lasers
- Atomic gas lasers
- Ion lasers
- Molecular lasers- the carbon dioxide laser
- Liquid dye lasers
- The free electron laser

coherence—well-defined frequency, phase, and polarization

can be concentrated into an area $\sim 4\times10^{-11}\ m^2$

cutting and welding in industry,
surgery cutting or cauterizing to stop bleeding, optical fiber conveying,
precise distance measurement,
holography, advertisement, bar code,
laser fusion, and laser weapons etc.

25.7 X ray

X ray: λ ~ nm. In 1901 Röntgen (1845–1923) won the first Nobel Prize

Bremstrahlung

Bremsung + Strahlung $\longrightarrow$ continuous X-ray spectrum

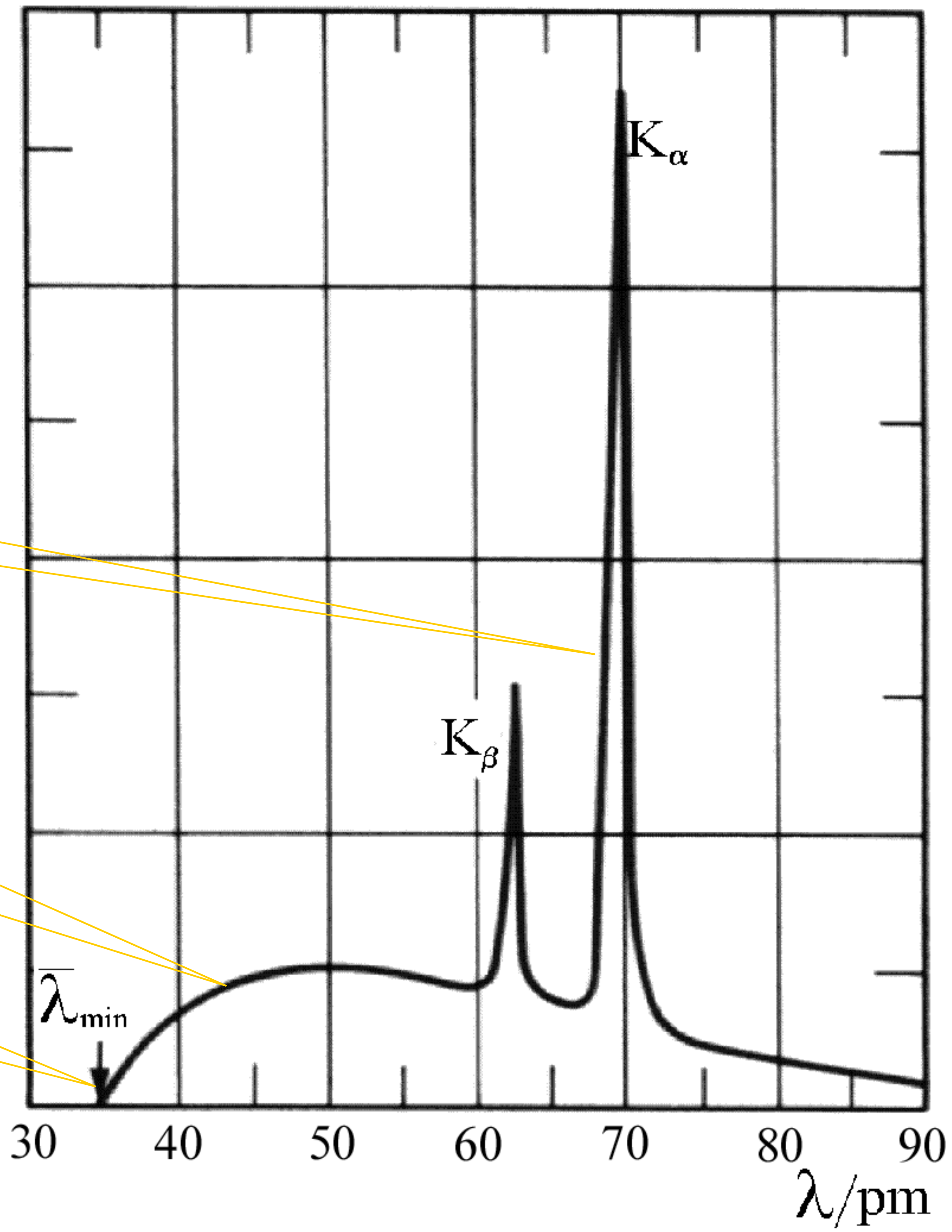
deceleration radiation

When a beam of 35keV electrons bombards molybdenum target

characteristic spectrum

continuous spectrum

minimum wavelength



cutoff-wavelength

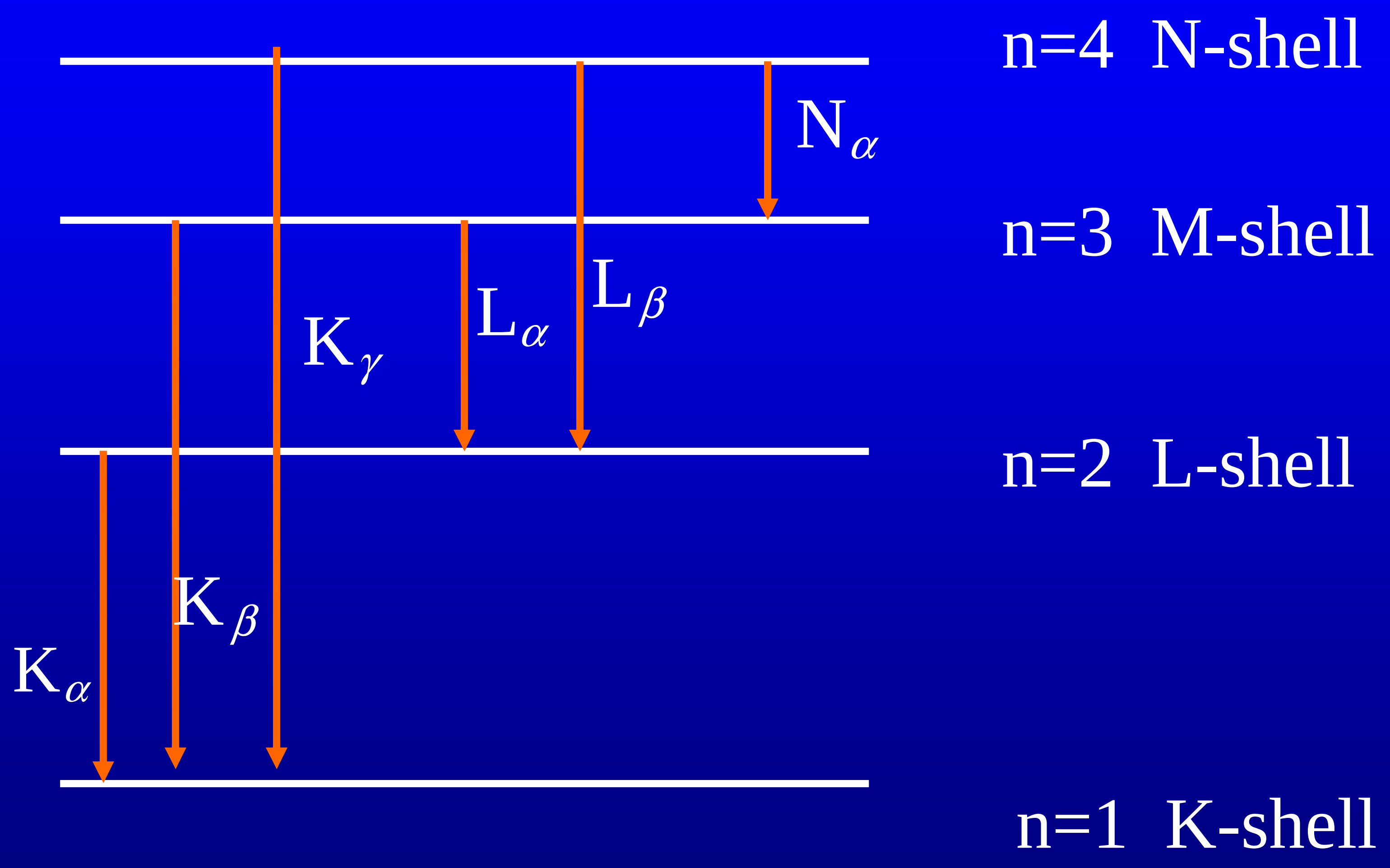
$$\lambda_{min} = \frac{hc}{E_k} = \frac{12\,400\,\mathrm{eV} \cdot \mathrm{\mathring{A}}}{35\,\mathrm{keV}} = 35.4\,\mathrm{pm}$$

All kinetic energy of the electron is changed into the energy of the phonon.

The relation between characteristic spectrum(discrete X-ray line spectra) and atomic energy levels

The inner electrons can be knocked off by the accelerated electrons. The outer electrons jump down and emit phonons of high frequency.

Discrete X-ray line spectra



An L shell electron is screened by two 1s electrons, and so the effective nuclear charge is Z_{eff}=Z–2.

When one of 1s electron is removed, Z_{eff}=Z–1.

Using the relation of 22.4, for K_α, we have

$$h\nu = hcR_\infty (Z-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$$

or

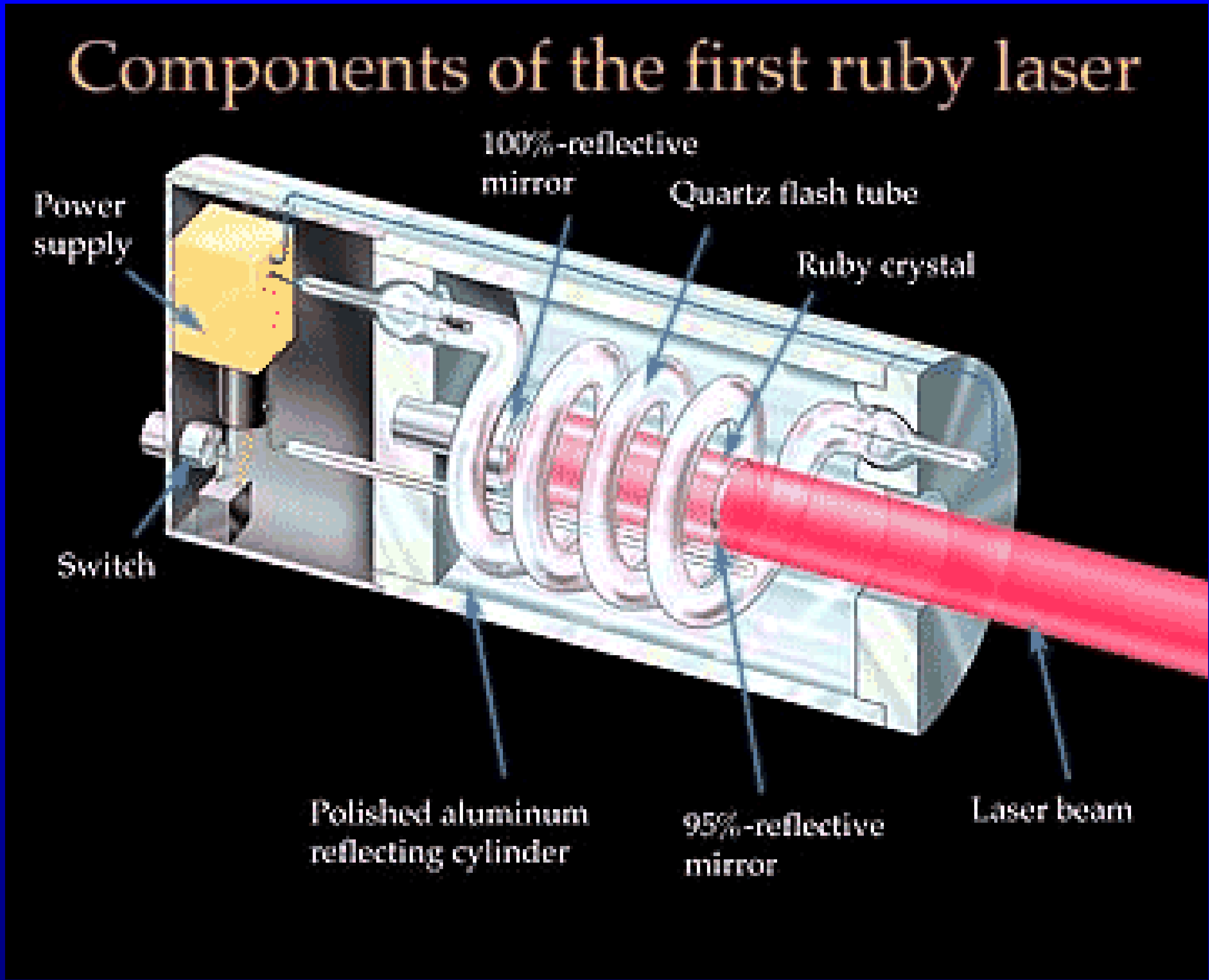
$$\nu = \frac{3cR_\infty}{4}(Z-1)^2.$$

Moseley law 1913 British scientist H. G. J. Moseley

Killed at Gallipoli in the First World War at age of 27.

It was early morning, August 10th, 1915. 30,000 fresh Turkish troops came down off Sari Bair hill on Gallipoli Peninsula. They fell upon the tired, badly positioned British. They destroyed them in furious hand-to-hand combat. Among the dead lay a 27-year-old lieutenant. He was Harry Moseley.

Assignment: 25.10,25.11


Components of the first ruby laser
100%-reflective mirror
Power supply
Quartz flash tube
Ruby crystal
Switch
Polished aluminum reflecting cylinder
95%-reflective mirror
Laser beam